Republic of Zambia

Ministry of Health

ZAMBIA NATIONAL SEXUALLY TRANSMITTED INFECTIONS TREATMENT GUIDELINES

December 2022

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Foreword

The burden of Sexually Transmitted Infections (STIs) remains very high, worldwide. The world health organisation (WHO) estimates are that one million new cases of curable STIs occur every day with over 376 million new cases reported in 2016. In Zambia, the incidence of STIs constituted 1.4% of outpatient attendees in 2021, and this trend is on the rise.

The WHO global health sector strategy on STIs, 2016–2021 aims to eliminate STIs as a public health threat by 2030. This can only be achieved by 1) preventing people from being infected, and 2) providing prompt and effective treatment and care for infected people to avoid further transmission of STIs to other people. Consequently, the Government of the Republic Zambia (GRZ) continues to expand efforts to strengthen STI case management and access to high quality services close to where people live using standardised interventions.

Zambia adopted the syndromic approach to management of STIs and developed treatment guidelines in 1995 which were only revised in 2006. The current review updates the 2006 treatment guidelines and is informed by data from the 2016 syndromic management and Gonococcal (GC) drug sensitivity study undertaken by the University of Zambia, School of Medicine (UNZA-SOM). In addition, an adaption was made from Centre for Disease Control (CDC), WHO 2021 guidelines and SADC STI guidelines.

Ideally, syndromic treatment guidelines must be reviewed and validated every 2 to 3 years.
It is notable that it took eleven years before the first guidelines could be revised, and it has been 16 years since the last (2006) revision. This slow pace of validation is of great concern as it likely impacts negatively on the validity of syndromic diagnoses and efficacy of treatments.

The root cause of these delays is two-fold:

1) logistical difficulties encountered in accessing research funds for STIs, other than HIV, and
2) the rudimentary nature of the system for routine laboratory testing and aetiological diagnosis

Going forward, the Ministry of Health (MoH) affirms to address these barriers by strengthening links (through the National Health Research Authority Council, NHRAC) with potential funders of STI research and, with local research institutions. Further, in order to improve the quality and utility of routinely collected data, the Ministry will continue to build human and laboratory capacity for biological sample collection, processing, and examination.

In this regard, capacity to perform rapid, point-of-care (POC) tests and molecular assays for common STIs will continue to be built into integrated routine care of STI clients and ensure commodity availability as well as security. This, together with diligent data collection and reporting through the Ministry's improved Monitoring and Evaluation system, will greatly enhance quality and reliability of routine data. I am also pleased to note that the review team has incorporated use of POCs and laboratory testing at facilities where these are available.

It is my sincere hope that these guidelines will prove to be a powerful weapon for health workers across all levels of healthcare provision in Zambia in the fight against STIs.

Hon. Sylvia T. Masebo, MP
MINISTER OF HEALTH

Process of Guidelines Development

This national guideline is informed by the 2016 GC validation study that was undertaken by the UNZA-SOM and data from the SADC regional framework and the World Health Organisation. The last national STI treatment guideline was developed in 2006 and has never been reviewed ever since. The results of the validation study which was conducted by UNZA-SOM delayed being released till October 2017.

With support from Global Fund, the first meeting to review the guidelines was held in March 2020 and the review team comprised members of the National STI Technical Working Group (STI-TWG) who are drawn from the Ministry of Health HQ, Provinces, Defence Forces, Police service, Prison service and the Private Sector amongst other stakeholders. Unfortunately, the activity which was scheduled to have been completed by October 2020 was delayed due to the outbreak of the COVID-19 Pandemic in March 2020.

However, the team reconvened in October 2021 then May 2022 to finalise the guidelines. It is expected that once the document is completed, there will be need for orientation of health care providers from all levels of health service delivery.

Acknowledgement

I acknowledge the Sexuality Transmitted Infections National Technical Working Group (STIs TWG) members who provided critical guidance early in the revision of these guidelines, including input regarding chapter topics and on ways to make the guidelines more useful across all the levels of care.

I also acknowledge the experts who provided guidance through their review of individual chapters. The revision would not have been possible without inputs from the STI validation study findings; I therefore also acknowledge the work of the STI validation study team in providing evidence for the common STIs and causative organisms in Zambia.

This revision has been supported over time from 2019 by the Global Fund, CDC-CoAg and MoH GRZ. The MoH expresses its gratitude to the following members of the National STI Technical Working Group (STIs TWG) for their dedication in ensuring the completion of the revised guidelines:

Brigadier General (RTD) Dr. Panganani Njobvu – Chair of the National STI Technical Working Group
Prof. Patrick Musonda, UNZA SoM
Dr. Dean Phiri, Global Fund PMU
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Permanent Secretary – Technical Services
MINISTRY OF HEALTH

List of Acronyms and Abbreviations

Acronym	Expansion
AIDS	Acquired Immune Deficiency Syndrome
BD	Twice Daily
BV	Bacterial Vaginosis
CDC	Centre for Disease Control
CSF	Cerebrospinal fluid
DHO	District Health Office
ELA	Enzyme-Linked Assay
GC	Gonococcal
GDS	Genital Discharge Syndrome
GRZ	Government of the Republic of Zambia
GUS	Genital Ulcers Syndrome
GV	Gardnerella Vaginalis
HIV	Human Immunodeficiency Virus
HMIS	Health Management Information System
HQ	Head Quarters
HSV	Herpes Simplex Virus
IgM-EIA	Enzyme Immunoassay for detection of Immunoglobulin M
IM	Intramuscular
IV	Intravenous
MB	Mycobacterium
MoH	Ministry of Health
NAATs	Nucleic Acid Amplification Tests
NGU	Non-Gonococcal Urethritis
Nocte	At Night
NTH	Ndola Teaching hospital
OD	Once Daily
PCR	Polymerase Chain Reaction
PO	Per Oral
POC	Point of Care
VMMC	Voluntary Medical Male Circumcision

Acronym	Expansion
cART	Combined Anti-Retroviral Therapy
RPR	Rapid Plasma Reagin
SADC	Southern African Development Community
STI	Sexually Transmitted Infection
TB	Tuberculosis
TDS	Three Times a Day
TPHA	Treponema Pallidum Hemagglutination Assay
TPPA	Treponema Pallidum Particle Agglutination
VDRL	Venereal Disease Research Laboratory
UDS	Urethral Discharge Syndrome
UNZA-SOM	University of Zambia – School of Medicine
UTH	University Teaching Hospital
VDS	Vaginal Discharge Syndrome
WHO	World Health Organisation
ZDHS	Zambia Demographic Health Survey
ZNPHI	Zambia National Public Health Institute
TCA	Trichloroacetic Acid

SECTION 1

SEXUALLY TRANSMITTED INFECTION SITUATION IN ZAMBIA

BACKGROUND

In Zambia, Sexually Transmitted Infections (STIs) have continued to be among the major causes of outpatient attendances both in the public and private health institutions. The most affected age groups are the sexually active population of between 15 – 25 years.

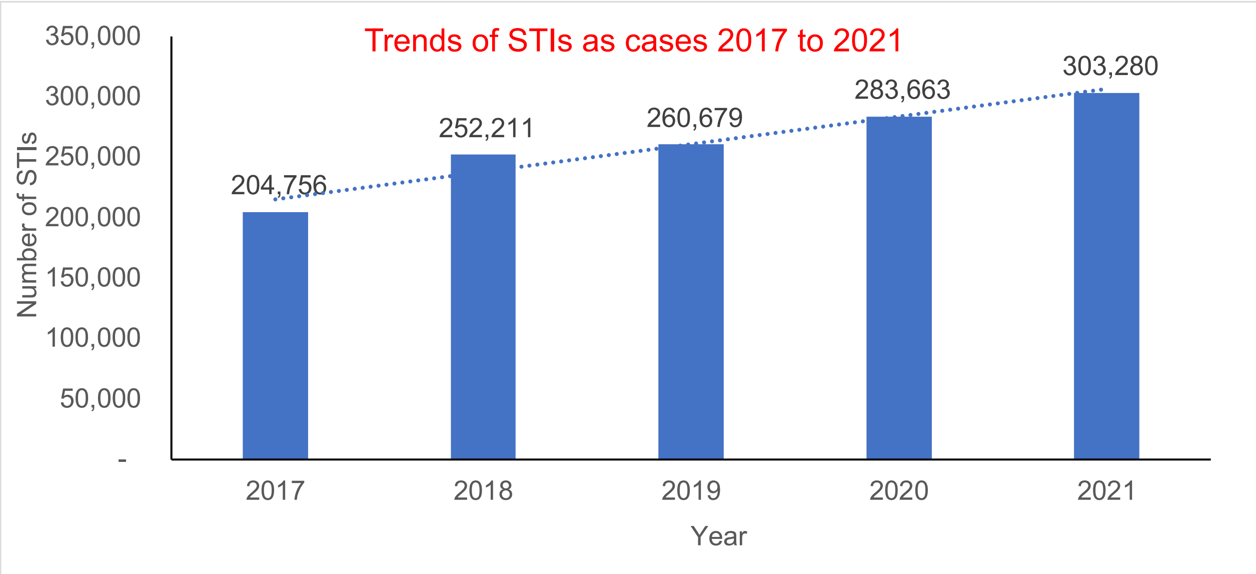


Figure 1: Trends of STI Outpatient Attendances in Zambia, 2017–2021

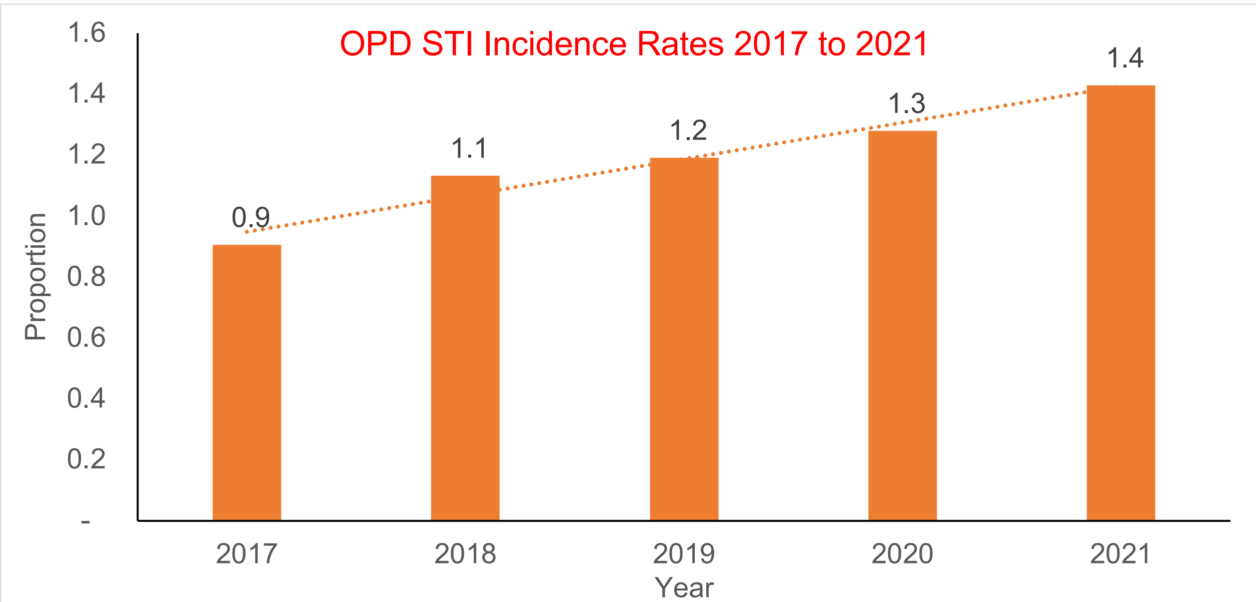


Figure 2: Proportion of STIs among OPD attendees by year, 2017–2021

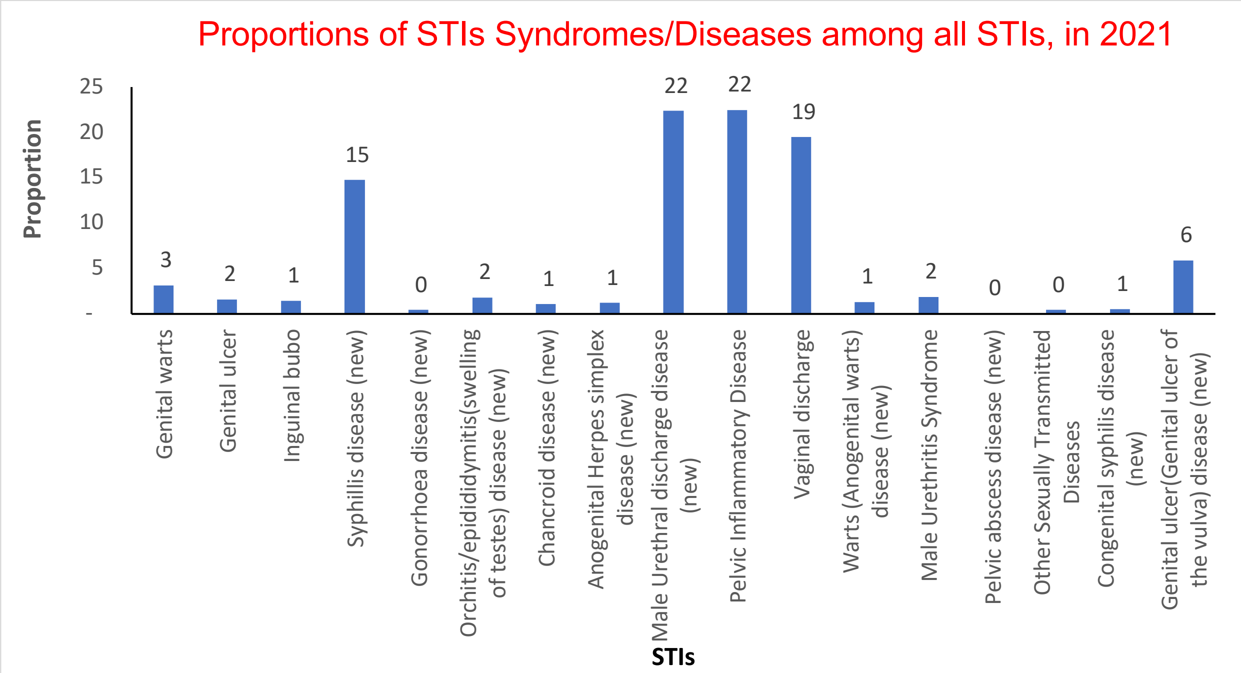


Figure 3: Proportions of STIs Syndromes/Diseases among all STIs recorded in 2021

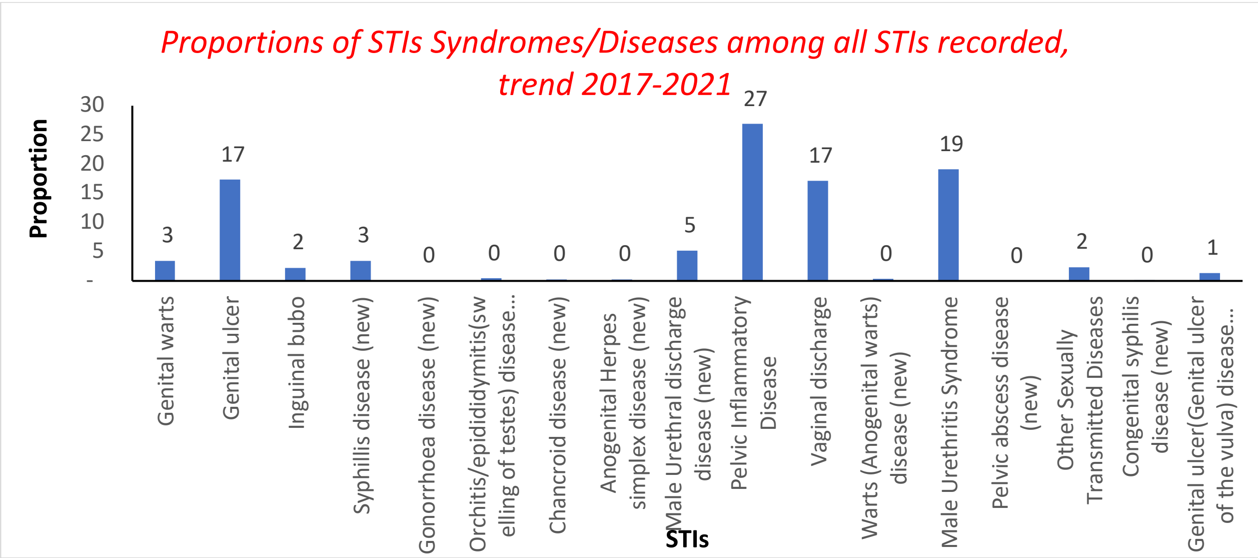


Figure 4: Proportions of STIs Syndromes/Diseases among all STIs recorded, 2017–2021

Source: **HMIS 2017-2021**

According to HMIS (2017-2021), the number of new sexually transmitted infections has been rising from 237,531 cases in 2017 to 503,325 cases in 2021. The commonest STI syndromes recorded from 2017 by frequency were lower abdominal pain syndrome, genital ulcers syndrome (GUS), vaginal discharge syndrome (VDS) and urethral discharge syndrome (UDS). The frequently affected age range was between 15 and 49 years. This also affirms the findings of the 2016 validation study which showed that genital discharges and genital ulcers were common in Zambia. In terms of causative agents isolated *Gardnerella vaginalis* accounted for 71%, *Ureaplasma urealyticum* 54%, Herpes simplex virus 2(HSV) 41% and *Trichomonas vaginalis* 14%. These findings coupled with the annual statistics show that STIs are still a problem of public health importance in the country and one of the main reasons for this occurrence include low condom utilisation. Among the 15- to 49-year-old respondents of the 2018 Zambia Demographic and Health Survey (ZDHS), only 35% of females and 54% of males reported using a condom when having sex with a person who was neither their spouse nor lived with them in the 12 months before the survey. The proportions were even lower among young people. Only 34% of females and 49% of males aged 15 to 24 years used a condom at the last high-risk sex with a non-regular partner.

Epidemiology of Sexually Transmitted Infections

In 2021 the World Health Organisation (WHO) estimated that globally, 1 million STIs are acquired daily. Based on prevalence data from 2009 to 2016, the WHO published a document in 2019 with estimated incident cases of chlamydia, gonorrhoea, syphilis and trichomoniasis of 376.4 million among people aged 15–49 years. There were 127.2 million new cases of chlamydia, 86.9 million new cases of gonorrhoea, 156 million new cases of trichomoniasis and 6.3 million new cases of Syphilis. Furthermore, data shows that the prevalence of some viral STIs is similarly high, with an estimated 417 million people infected with Herpes Simplex Virus type 2 (HSV-2) and about 291 million women with Human Papillomavirus (HPV) at any point in time. In recent years, evidence has accumulated suggesting that most cases of genital ulcer syndrome in Sub-Saharan Africa are due to viral and not bacterial infections. Findings from a recent study in Botswana suggest a reduction in the proportion of ulcers due to bacterial causes such as syphilis and chancroid, and an increase in the proportion of ulcers due to HSV. HSV has been attributed to 30-50% of GUS cases, as shown from studies in Malawi, Namibia, Eswatini (Swaziland) and South Africa, that indicate the emergence of viral causes (HSV-2) in the pathogenesis of GUS. The change in epidemiology may be attributed to the introduction of syndromic management of STIs. The most common pathogens in UDS patients in the SADC region are *Neisseria gonorrhoeae*, *Chlamydia trachomatis* and *Trichomonas vaginalis*. *N. gonorrhoeae* is implicated in 50-60% of cases, *C. trachomatis* in 20-30% of UDS and *T. vaginalis* has also emerged as a frequent pathogen in 5-20% of cases. Similarly, trichomoniasis and bacterial vaginosis are common causes of VDS amongst women. Drug resistant forms of *N. gonorrhoeae* have been evolving in the SADC region. Studies in Malawi, Mozambique, South Africa, and Tanzania have assessed the antimicrobial susceptibility profile of gonococcal isolates in the respective geographical locations. Studies from

South Africa illustrated high levels of resistance to penicillin and tetracyclines. Ciprofloxacin (quinolone) resistance ranged from 7% to 44%. Overall, the gonococcal isolates remained susceptible to Ceftriaxone, Cefixime, and Spectinomycin.

In Zambia, both GUS and GDS remain major causes of STIs seen by clinicians. In the validation study of 2016, these are seen in equal numbers in both males and females and have continued to have high prevalence. This study revealed that the most common organisms responsible for genital discharge syndrome (GDS) are *G. vaginalis*, *N. gonorrhoeae* and *U. urealyticum*. Usually, the pathogens responsible for GDS are *N. gonorrhoeae*, *C. trachomatis*, *M. genitalium*, *U. urealyticum* and *T. vaginalis*. Inguinal buboes are on a drastic decline across the Country and this observation is in line with what is happening elsewhere in the region. There are however significant differences between provinces for all the syndromes (2016 Zambia Validation Study) see table below:

1.1.1 Zambia 2016 Validation Study – Identified Causative Pathogens

Table 1: Pathogens associated with Genital/Urethral Discharge Syndrome

Pathogen	Frequency
N. gonorrhoeae	Male 70% Female 64%
C. trachomatis	Male 16% Female 21%
T. vaginalis	Male 9% Female 47%
M. genitalia	Male 4% Female 21%
U. urealyticum	Male 50% Female 66%
G. vaginalis	Male 64% Female 73%
Source: Ministry of Health (2016). Draft Report on Validation Study of Syndromic Management of Sexually Transmitted Infections in Zambia. [Unpublished Manuscript]	

Table 2: Pathogens associated with Implicated in Genital Ulcer Syndrome

Pathogen	Frequency
HSV-1	2%
HSV-2	41%
T. pallidum	6%
No tests done for LGV, Donovanosis or Chancroid	
Source: Ministry of Health (2016). Draft Report on Validation Study of Syndromic Management of Sexually Transmitted Infections in Zambia. [Unpublished Manuscript]	

The molecular identification of causative pathogens for the various STI syndromes has revealed multiple infections in almost all the specimens collected. New organisms that were not considered during the initiation of the syndromic approach to manage STIs have now emerged and may account for the reported non-response to treatment using the current algorithms. *Mycoplasma genitalium* and *Gardnerella vaginalis* are examples of this finding. With availability of molecular assays, identification of such organisms will be more accurate. *M. genitalium* is an extremely slow-growing and fastidious bacterium causing a well-established STI and the etiological agent of a number of syndromes.

Status of The National STI Control Program

The Zambian Government's focus on HIV and AIDS just like other SADC member states is broadened and this is attributed to the recognition of the importance of STIs as one of the factors that have been fuelling the AIDS epidemic. The Government's investment in STI control has increased steadily especially with the increasing support that the country has been receiving from Global Fund and PEPFAR though still much needs to be done. This evidenced through the Government's creation of the National AIDS Council which is mandated to champion the prevention and control of both HIV and AIDS as well as Sexually Transmitted Infections and also the unit within the Ministry of Health that oversees these operations. To this effect, National HIV/AIDS/STI strategic plans have been developed and operationalized over the years at all levels of service provision.

It is also important to note that Zambia first developed the National STI Syndromic management guidelines in 1995 and had not been revised until 2006. The development of the 2006 National

STI Syndromic guidelines was done after the findings of 2002 study by Luo et al. After the guidelines were developed, a team of experts also went ahead and developed training materials/modules which were used to train health care providers in the public as well as the private sectors. This included staff from the Defence Forces, Prison services, Police services and Church health organizations amongst others. To improve health care delivery, the nursing and clinical officer tutors were trained and the STI syndromic management course found itself in the preservice nursing and clinical officer training curriculum. Over 4000 health care providers were trained.

Over the years, the government has strengthened other HIV/STI prevention and control interventions which include the Condom Programming, VMMC and cART. All these signify the high political commitment that has been shown towards the prevention and control of STIs.

Current Challenges

Despite making inroads in the prevention and control of STIs through a robust countrywide approach to the promotion and distribution of condoms through the CONDOMIZE Campaign programme, the country still faces several challenges.

1.1.2 Key Service Delivery Challenges

a) Service Delivery

- Inadequate strategies to reach the vulnerable and most-at-risk populations (MARPs)

b) Partner management

- One of the main challenges in the management and prevention of STIs lies in the tracing and treatment of contacts to index cases. There is urgent need to devise workable solutions to this and includes making available adequate partner contact slips
- In order to enhance efficiency in various programs, there is need to promote integration

c) Human Resource

- The recent recruitment and deployment of over 11,000 healthcare workers will greatly help address the gaps created by inadequate qualified human resource in the past

d) Leadership and Governance

- In order to ensure smooth running of the STI programs, there is need to appoint focal point persons at both district and provincial levels to champion the STI agenda
- The private sector needs to be engaged at all levels of care to ensure utilization of standardized national guidelines in the management of STIs

e) Commodity availability (drugs and other medical supplies)

- In order to ensure consistent availability of medicines and other medical supplies, there is need to prioritise the availability of Health Center Kits since these contain most of the basic supplies including Benzathine penicillin and Doxycycline, e.t.c

f) Data management

- In order to assure quality data being generated, essential data collecting and reporting tools should be consistently available such as STI registers, OPD and IPD registers, tally sheets and contact slips. This should be accompanied by training/orientation or mentorship of staff

SECTION 2

THE NEW FRAMEWORK FOR STI MANAGEMENT IN ZAMBIA

RATIONALE

According to the WHO, effective management of STIs by using proven efficacious drugs is one of the cornerstones of STI prevention and control. This prevents the development of complications and sequelae, decreases the spread of STIs in the community and offers a unique opportunity for targeted education about HIV/STI prevention.

Therefore, prompt and effective treatment of STI clients at their first encounter with a health care provider is an important public health measure as it reduces the duration of time patients remain infectious, and the transmission of the infections. One of the fundamental ways to achieving this is the promotion of standardized and periodic revision/updating of national case management guidelines to ensure adequate therapy at all levels of health service delivery. Such standardized and updated guidelines will not only facilitate the training and supervision of health care providers but will also play a major role in preventing and/or delaying the development of resistance to antimicrobial agents and is an important factor in rational drug procurement. To attain this, the Zambian National STI Treatment Guidelines have incorporated other approaches to treatment, particularly the Aetiological and the clinical approaches.

CRITERIA USED FOR SELECTING RECOMMENDED DRUGS IN THESE GUIDELINES

The treatments recommended in these guidelines have been selected according to the following criteria:

1. High efficacy, effecting cure in at least 95% or more of cases.
2. Approved for use in Zambia
3. Cost-effective
4. Safe, with acceptable toxicity and tolerance
5. Likely sustained microbial susceptibility: i.e., organism resistance unlikely to develop or likely to be delayed
6. Easily administered and convenient: i.e., single dose or oral administration whenever possible

Recommended Drugs and other Medical Supplies for STI Management

The following is the list of recommended drugs and other medical supplies in the management of STIs in Zambia:

Table 3: Recommended Drugs and other Medical Supplies for STI Management

Antibiotics	***Antivirals***	**Antiprotozoal**	**Antifungals**	**Topical**	**Other supplies**
Cefixime	Acyclovir	Metronidazole	Fluconazole	Podophyllin	Male/female condoms
Erythromycin	Famciclovir	Tinidazole	Clotrimazole cream	Silver nitrate	Speculums
Azithromycin	Valacyclovir	Ivermectin	Nystatin pessaries	Trichloroacetic acid (TCA)/ Bichloroacetic acid	Examination lamps
Doxycycline				Imiquimod	Cryotherapy devices/equipment
Benzathine penicillin				Benzyl Benzoate	Portable Diathermy
Erythromycin				Permethrin	Swabs
Ceftriaxone				Lindane Cream	PCR machines
Co-trimoxazole				Clindamycin	Suspensory bandages
				20% and 40% Lignocaine Cream	Laboratory reagents
				Vaseline Jelly	Vaccines (HPV and Hepatitis B)
				5% Imiquimod cream	PH paper Strips

Recommended Treatment Regimens for Syndromic Treatment of STIs

The following is the list of recommended treatment regimens for syndromic treatment of STIs:

Table 4: Recommended Treatment Regimens for Syndromic Treatment of STIs

STI Syndrome	Recommended Actions	Preferred drugs	Alternative drugs
Vaginal Discharge	Treat for gonorrhoea	Ceftriaxone 500mg IM stat (to treat gonococcal infection) Children and adolescents (≤17 years) Ceftriaxone 25-50mg/kg body weight IM stat If ≥45kg, use adult dose	Cefixime 400mg PO stat Children/adolescents <17 years Cefixime 8mg/kg body weight PO stat **Note**: If patient allergic to Cephalosporin use Gentamycin 240mg IV stat
	Treat for chlamydia	Doxycycline 100mg PO BD for 7 days	Azithromycin 1g PO stat **OR** Erythromycin 500mg PO QID 7 days **(in pregnant or lactating woman)**
	Treat for Vaginal candidiasis	Fluconazole 150mg PO stat	Clotrimazole vaginal pessaries 200mg OD for 3 days **OR** Miconazole vaginal pessaries 200mg for 3 days **OR** Nystatin pessary 100,000units Nocte for 14 days

Table 4: Recommended Treatment Regimens for Syndromic Treatment of STIs

STI Syndrome	Recommended Actions	Preferred drugs	Alternative drugs
	Treat for Bacterial vaginosis	Metronidazole 400mg PO BD for 7 days **OR** Metronidazole gel 0.75% 5g (full applicator) intravaginally OD for 5 days **OR** Clindamycin cream 2% one full applicator (5g) intravaginally OD for 7 days	Clindamycin 300mg PO BD for 7 days **OR** Tinidazole 2g PO BD for 3 days **OR** Tinidazole 1g PO OD for 5 days
	Treat for trichomoniasis	Metronidazole 2g PO stat **OR** Metronidazole 400mg PO BD for 7 days For children ≤17 years 5mg/kg PO TDS for 7 days	Tinidazole 2g PO stat
Genital Ulcer	Treat for syphilis	Benzathine penicillin 2.4 MU IM stat as a single injection split as 1.2 MU given in each buttock	Erythromycin 500mg PO QID for 14 days **(in pregnant or lactating woman)**
	Treat for chancroid	**PLUS** Azithromycin 1g PO stat	

Table 4: Recommended Treatment Regimens for Syndromic Treatment of STIs

STI Syndrome	Recommended Actions	Preferred drugs	Alternative drugs
	Treat for Lymphogranuloma venerium (LGV)	**PLUS** Doxycycline 100mg PO BD for 21 days	
	Treat for Herpes simplex	**PLUS** Acyclovir 400mg PO TDS for 7 days	
Urethral Discharge Syndrome	Treat for gonorrhoea Treat for chlamydia If urethral discharge persists after 7 days despite adequate treatment and no history of re-exposure to STI then treat with	Ceftriaxone 500mg IM stat **PLUS** Doxycycline 100mg PO BD for 7 days, Metronidazole 2g PO stat (to treat Trichomonas vaginalis) **PLUS** Doxycycline 100mg PO BD for 7 days **followed by** Azithromycin 1g PO stat, then 500mg PO OD for 3 days (to treat M. genitalium)	Cefixime 400mg PO stat Azithromycin 1g PO Stat **OR** Erythromycin 500mg PO QID for 7 days
Inguinal Bubo†	Aspirate FLUCTUANT bubo with a large bore needle through normal skin **Treat for chancroid**	Azithromycin 1g PO stat **PLUS** Doxycycline 100mg PO BD for 21 days	Ceftriaxone 500mg IM stat **OR** Cefixime 400mg PO stat If pregnant or lactating woman use

Table 4: Recommended Treatment Regimens for Syndromic Treatment of STIs

STI Syndrome	Recommended Actions	Preferred drugs	Alternative drugs
	Treat for LGV		Erythromycin 500mg PO QID 21 days (**Note**: Do not use Doxycycline)
Female Lower Abdominal Pain (PID) Outpatient	Treat for gonorrhoea, chlamydia and anaerobic bacteria at the same time	Ceftriaxone, 500mg IM stat **PLUS** Doxycycline 100 mg PO BD for 14 days **PLUS** Metronidazole 400mg PO TDS 14 days **OR** **Metronidazole** 2g PO stat (to treat anaerobic bacteria)	Cefixime 400mg PO stat **PLUS** Erythromycin 500mg PO QID for 14 days
Female Lower Abdominal Pain (PID) Inpatient	Treat for gonorrhoea, chlamydia, and Anaerobic bacteria at the same time	Ceftriaxone 1g IV stat **PLUS** Doxycycline 100mg PO BD for 14 days **PLUS** Metronidazole 400mg PO TDS for 14 days **OR** Metronidazole 500mg IV BD for 7 days **Note: Switch to oral treatment upon clinical improvement**	Cefixime 400mg PO stat **PLUS** Erythromycin 500mg PO QID for 14 days

Table 4: Recommended Treatment Regimens for Syndromic Treatment of STIs

STI Syndrome	Recommended Actions	Preferred drugs	Alternative drugs
		and continue up to 14 days	
Genital Growth	Treat for Condylomata acuminata (ano-genital warts)	Podophyllin 25% tincture. Apply topically weekly up to 12 weeks (protect surrounding normal skin with Vaseline jelly before application by health care provider) Imiquimod cream 5% applied at bedtime 3 times a week up to a duration of 8-12 weeks	Cauterization, Trichloroacetic Acid (weekly), Cryotherapy (fortnightly) Surgical excision of warts for isolated lesions
	Treat for Condylomata lata	Benzathine Penicillin 2.4 MU IM stat	
Scrotal Swelling	Treat for Gonorrhoea	Ceftriaxone 500mg IM stat **PLUS**	Cefixime 400mg PO stat
	Treat for Chlamydia	Doxycycline 100mg PO BD for 14 days	Azithromycin 1g PO weekly for 2 weeks (Total of 2 doses) **OR** Erythromycin 500mg PO QID for 14 days

Table 4: Recommended Treatment Regimens for Syndromic Treatment of STIs

STI Syndrome	Recommended Actions	Preferred drugs	Alternative drugs
Neonatal Conjunctivitis	Treat for gonorrhoea and chlamydia	Ceftriaxone 25 – 50mg/kg body weight (max. 125mg) IM stat **PLUS** Erythromycin syrup, 50mg/kg body weight PO daily in 4 divided doses for 14 days **PLUS** Saline lavage of eyes	Cefotaxime 100mg per kg/body weight IV/IM stat

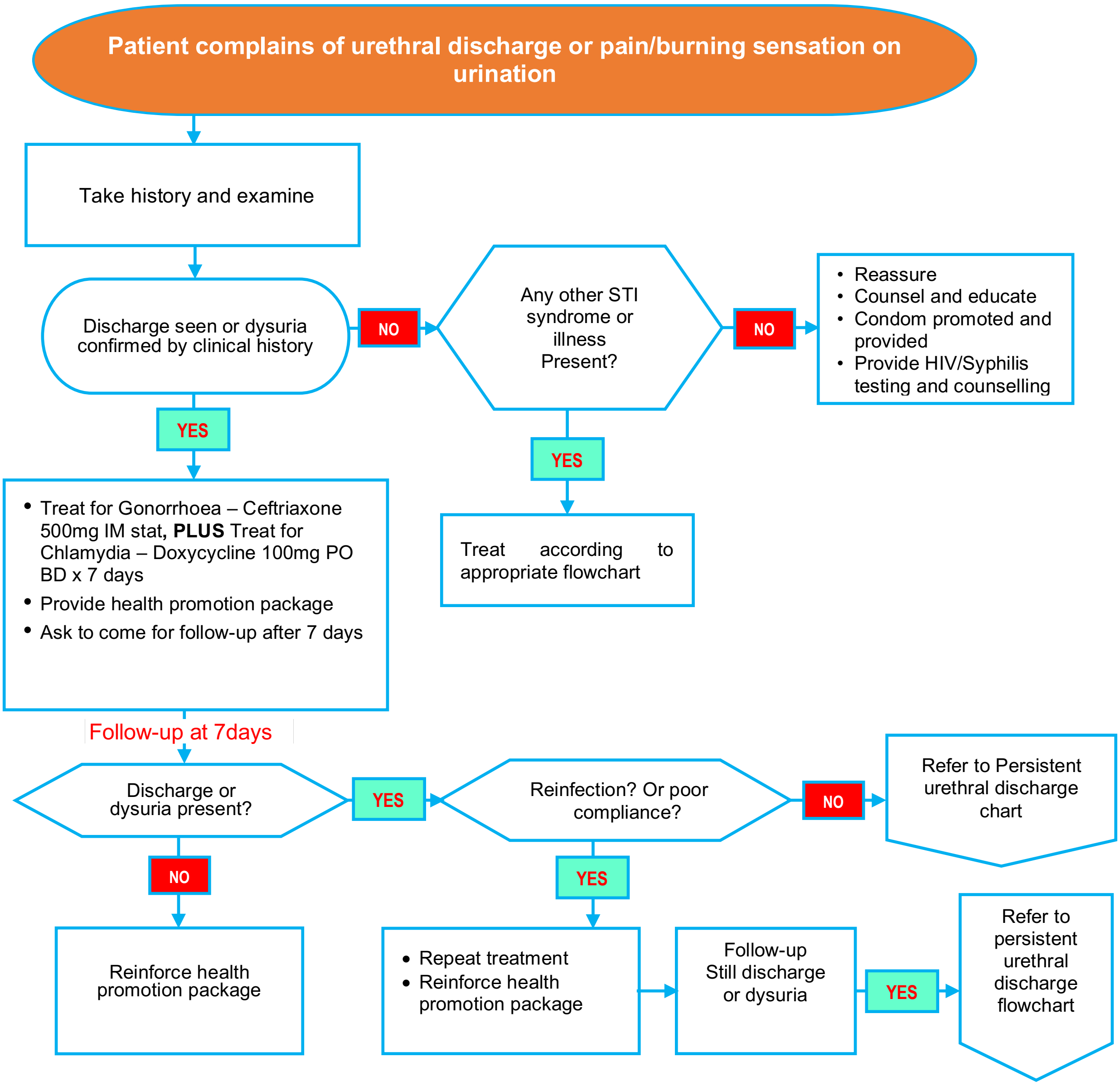
* **Risk Assessment.** The vaginal discharge algorithm is not very sensitive for predicting presence of cervical infection (Gonorrhoea and Chlamydia). Speculum examination improves its diagnostic utility. However, presence of certain risk factors increases the sensitivity and specificity of the algorithm for predicting cervicitis. Routine use of risk assessment is therefore recommended in all cases of vaginal discharge where speculum examination is not available or feasible. Consider risk assessment to be positive if the client is sexually active and has one or more of the following:

1. Has engaged in sex with multiple partners in last three months
2. Has had a new sex partner in the last three months
3. Has a current partner with an STI
4. Has a history of an inappropriately treated STI
5. Is a victim of sexual assault

† **Inguinal bubo** accompanied **with genital ulcer(s)** should receive treatment as for genital ulcer disease

Flowcharts For Syndromic Management Of Sexually Transmitted Infections

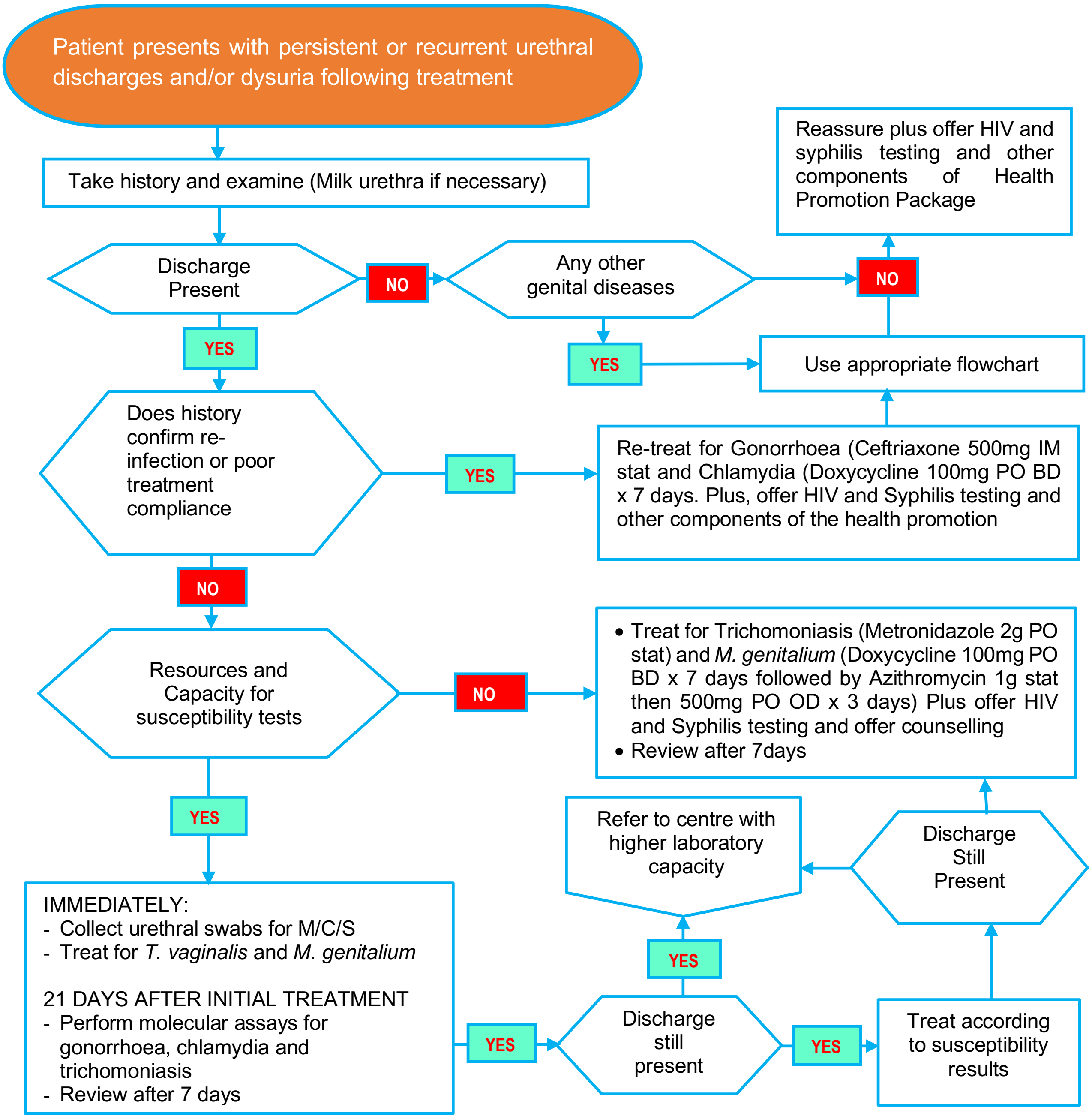
Urethral Discharge Syndromic Flowchart



HEALTH PROMOTION FOR ALL PATIENTS	
1) Educate, ensure compliance and counsel 2) Promote abstinence during the course of treatment 3) Promote and demonstrate condoms use, and provide condoms 4) Stress the importance of partner treatment and issue one notification slip for each sexual partner, follow up partner treatment during review visits	5) Provide HIV and syphilis testing, for negative test results repeat after three months 6) Offer male circumcision to uncircumcised partner/patient 7) Provide cervical cancer screening for female partner if indicated
TREATMENT	
PREFERRED	**ALTERNATIVE**
Ceftriaxone 500mg IM stat (Gonococcal)	Cefixime 400mg PO stat
Azithromycin 1g PO weekly for 2weeks (Chlamydia)	Doxycycline 100mg PO BD for 7 days or Erythromycin 500mg PO QID for 7 days
Metronidazole 2g PO stat	

Figure 5: Urethral Discharge Syndrome Flowchart

RECURRENT/PERSISTENT MALE URETHRAL DISCHARGE SYNDROME FLOWCHART

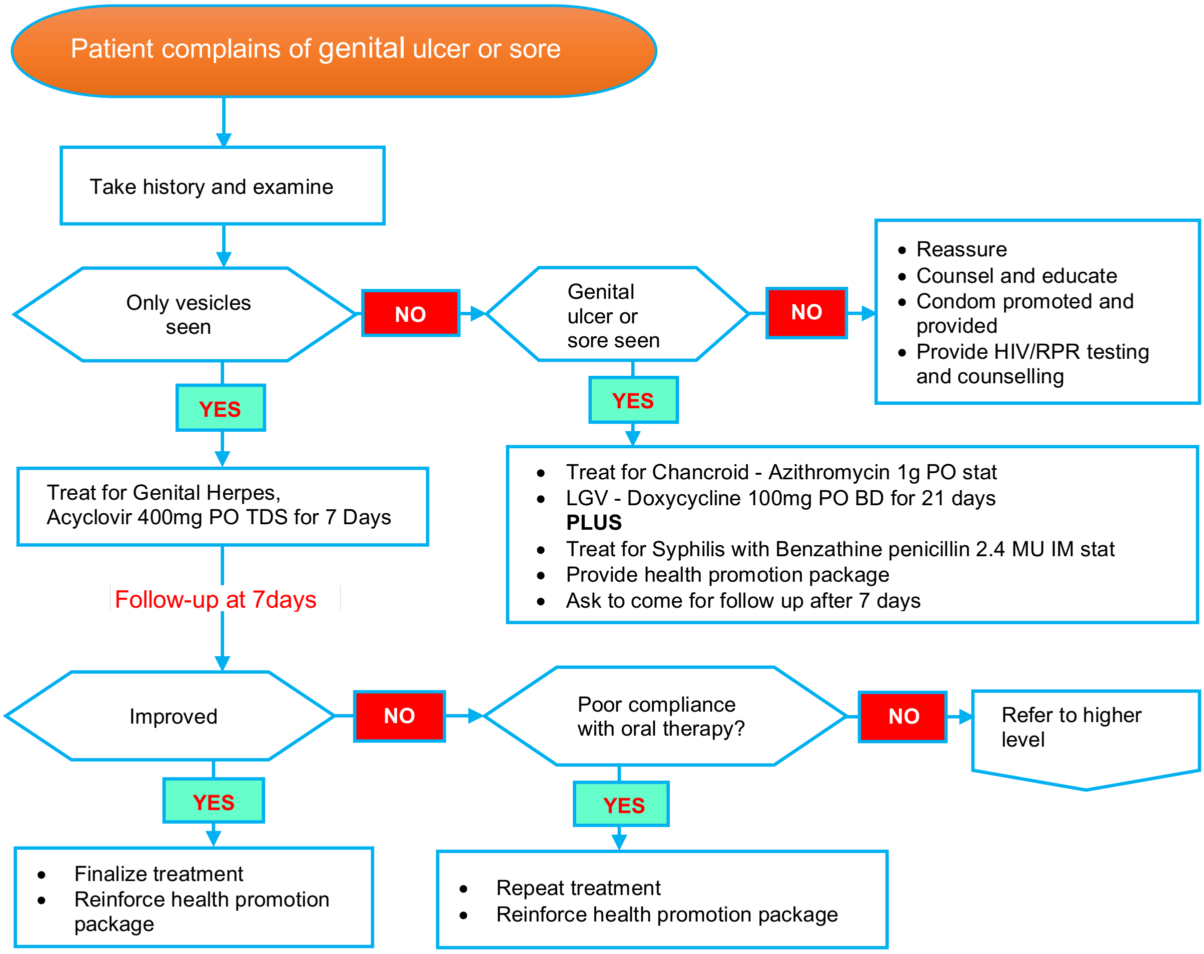


HEALTH PROMOTION FOR ALL PATIENTS

1) Educate, ensure compliance and counsel
2) Promote abstinence during the course of treatment
3) Promote and demonstrate condom use, and provide condoms
4) Stress the importance of partner treatment and issue one notification slip for each sexual partner, follow up partner treatment during review visits
5) Provide HIV and syphilis testing, for negative test results repeat after three months
6) Offer male circumcision to uncircumcised partner/patient
7) Provide cervical cancer screening for female partner if indicated

Figure 6: Recurrent/Persistent Urethral Discharge Syndrome Flowchart

Genital Ulcer Syndrome Flowchart

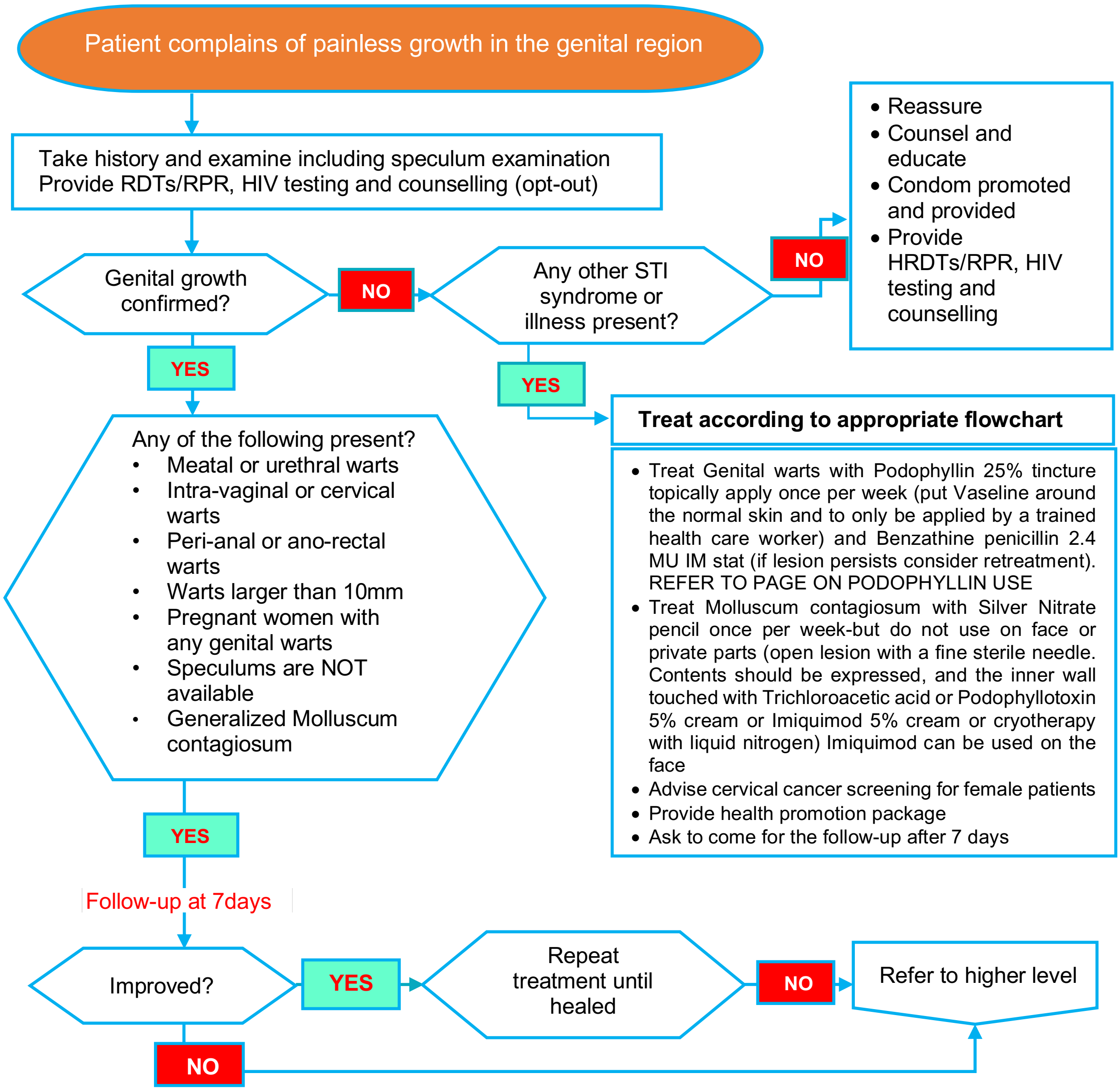


HEALTH PROMOTION FOR ALL PATIENTS	
1) Educate, ensure compliance and counsel 2) Promote abstinence during the course of treatment 3) Promote and demonstrate condom use, and provide condoms 4) Stress the importance of partner treatment and issue one notification slip to each sexual partner, follow up partner treatment during review visits	5) Provide HIV and syphilis testing, for negative test results repeat after three months 6) Offer male circumcision to uncircumcised partner/patient 7) Provide cervical cancer screening for female partner if indicated

TREATMENT	
PREFERRED	**ALTERNATIVE**
Benzathine penicillin 2.4 MU IM stat	
Azithromycin 1g PO stat	Erythromycin 500mg PO QID for 14 days
Acyclovir 400mg PO TDS for 7days	

Figure 7: Genital Ulcer Syndrome Flowchart

GENITAL GROWTH SYNDROME FLOWCHART



HEALTH PROMOTION FOR ALL PATIENTS	
1) Educate, ensure compliance and counsel 2) Promote abstinence during the course of treatment 3) Promote and demonstrate condom use, and provide condoms 4) Stress the importance of partner treatment and issue one notification slip to each sexual partner, follow up partner treatment during review visits	5) Provide HIV and syphilis testing, for negative test results repeat after three months 6) Offer male circumcision to uncircumcised partner/patient 7) Provide cervical cancer screening for female partner if indicated

TREATMENT	
PREFERRED	**ALTERNATIVE**
Podophyllin 25% tincture (genital warts)	Cauterization, cryotherapy, Phenol application, Trichloroacetic acid
Benzathine penicillin 2.4 MU IM stat (Syphilis)	
Silver Nitrate Pencil (Molluscum contagiosum)	Cauterization, cryotherapy, Phenol application

Figure 8: Genital Growth Syndrome Flowchart

GENITAL GROWTH SYNDROME FLOWCHART

Genital warts (Condylomata acuminatum) are non-painful, cauliflower like growths in the genital area.

They are caused by a Human Papilloma Virus (HPV). They should be differentiated from Condylomata lata, a manifestation of secondary syphilis, which usually has other manifestation such as a macular-papular non-itchy skin rash on the body, palms and soles of feet. In all cases of Condylomata lata, the rapid regain test (RPR) is strongly reactive

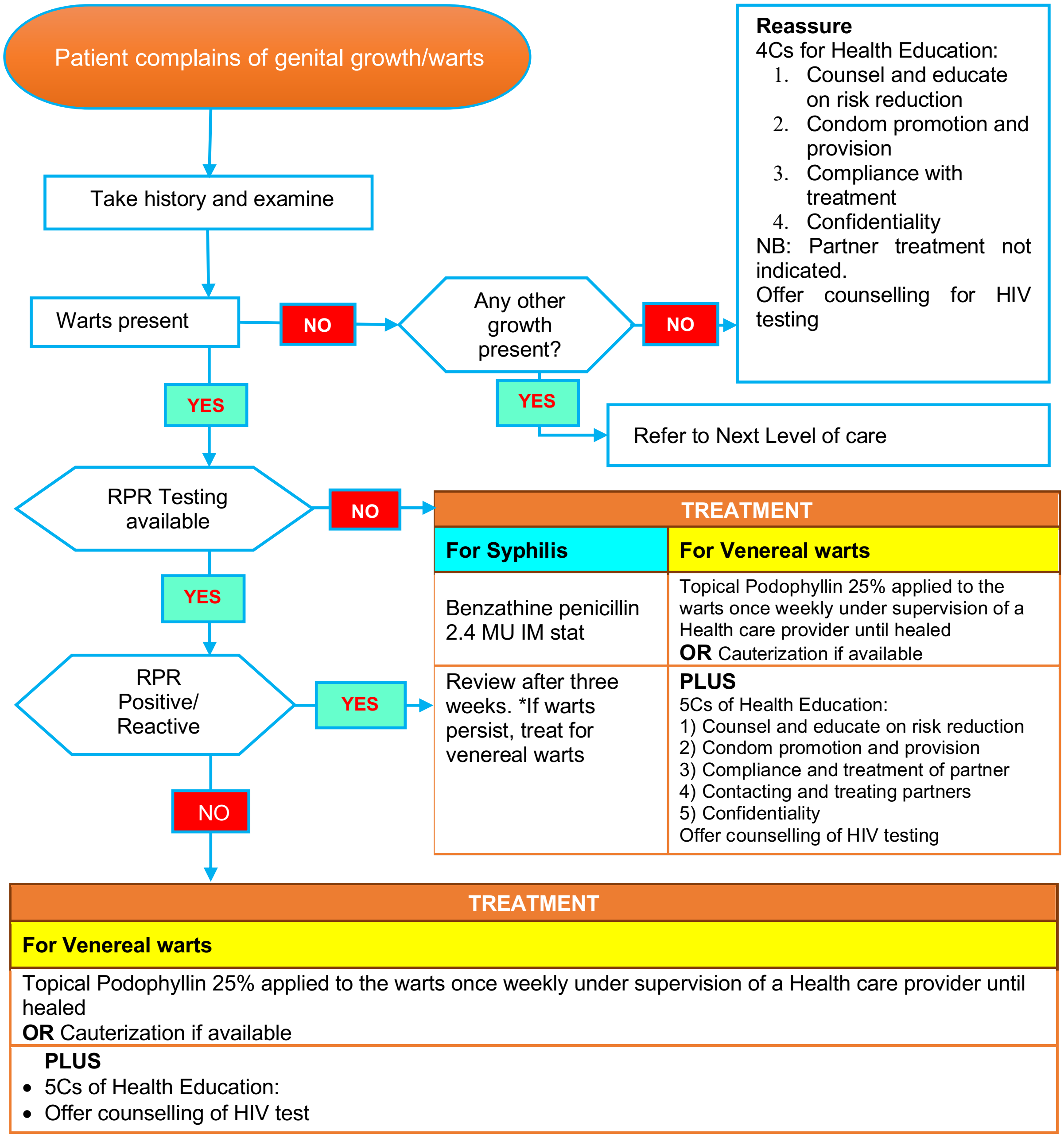


Figure 9: Genital Growth Syndrome Flowchart

VAGINAL DISCHARGE SYNDROME FLOWCHART

Patient complains of vaginal discharge, Vulva itching and/burning

Take history and examine.
- Assess risk *
- Rule out pregnancy (History/LMP/Scan/pregnancy test)

SEXUALY ACTIVE RISK ASSESSMENT

Consider risk assessment in sexually active clients with at least one of the other risk factors listed below:

- Multiple sexual partners in the previous 3 months
- Having a new partner in the previous 3 months
- Having a current partner with an STI
- Victim of sexual assault
- Inappropriately treated STIs

Lower abdominal tenderness or cervical motion tenderness present?

YES → Use flowchart for lower abdominal pain

NO

Cervical muco-pus/erosions or positive risk assessment?*

YES

TREATMENT for Gonococci, Chlamydia and Trichomoniasis	
Drug	**Condition**
Ceftriaxone 500mg IM stat **PLUS**	Gonorrhoea
Azithromycin 1g orally stat **PLUS**	Chlamydia
Metronidazole 2g stat	Trichomoniasis
Review after 7 days Health promotion package	
Treat cervicitis and ask client to come for cacx screening in 7 days	

If no improvement, refer to higher level for management. If no improvement, refer to higher level for management. Refer to evidence coming up in new studies for the single dose component

NO

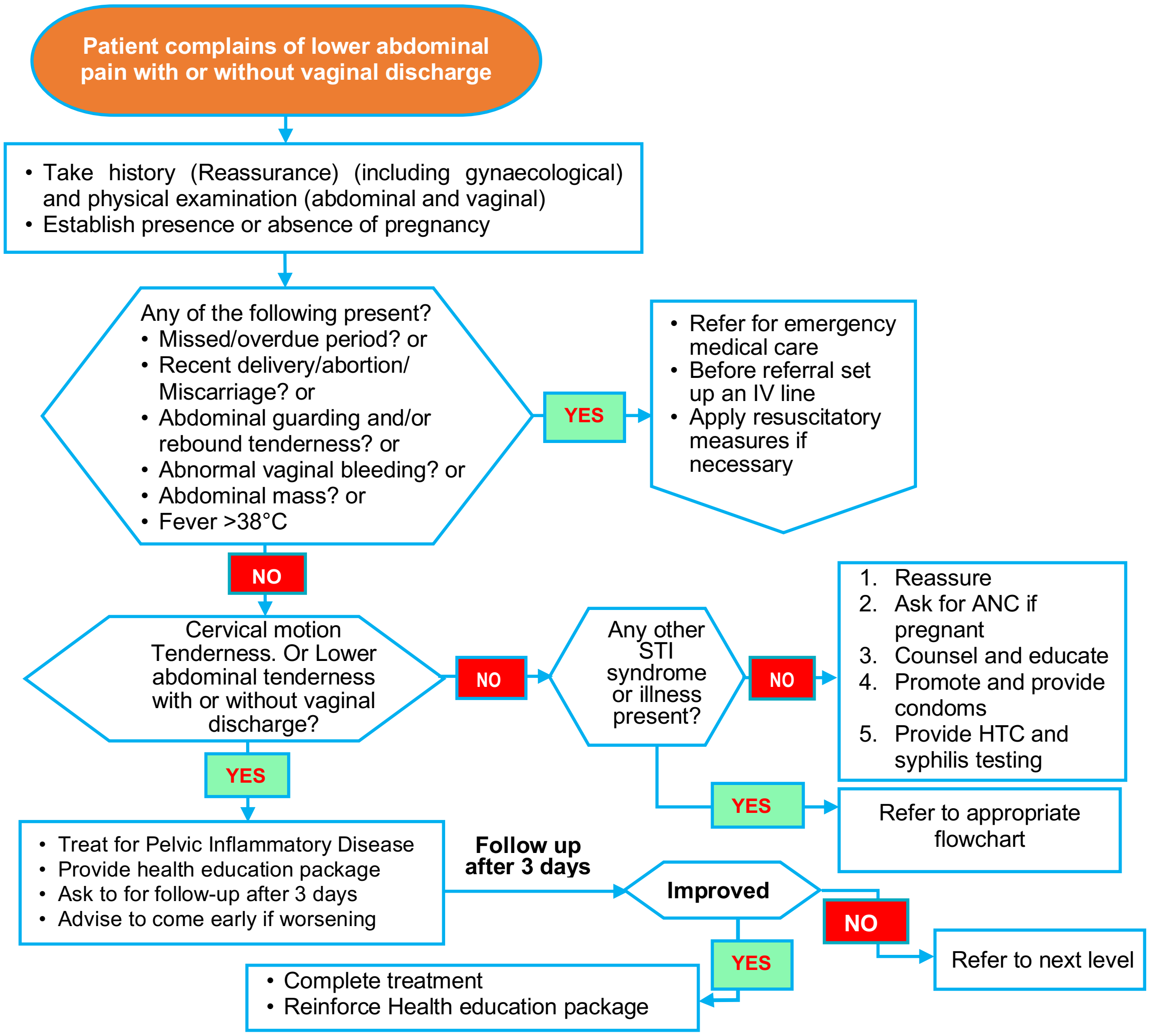
TREATMENT for Gonococci, Chlamydia and Trichomoniasis	
Drug	**Condition**
Metronidazole 2g stat orally plus	Bacteria vaginitis and Trichomoniasis
Fluconazole 200mg orally stat (Avoid in the first trimester of pregnancy)	Candidiasis
Health promotion package	

HEALTH PROMOTION FOR ALL PATIENTS

1) Educate, ensure compliance and counsel
2) Promote abstinence during the course of treatment
3) Promote and demonstrate condom use, and provide condoms
4) Stress the importance of partner treatment and issue one notification slip to each sexual partner, follow up partner treatment during review visits
5) Provide HIV and syphilis testing, for negative test results repeat after 3 months
6) Offer male circumcision to uncircumcised partner/patient
7) Provide cervical cancer screening for female partner if indicated

Figure 10: Vaginal Discharge Syndrome Flowchart

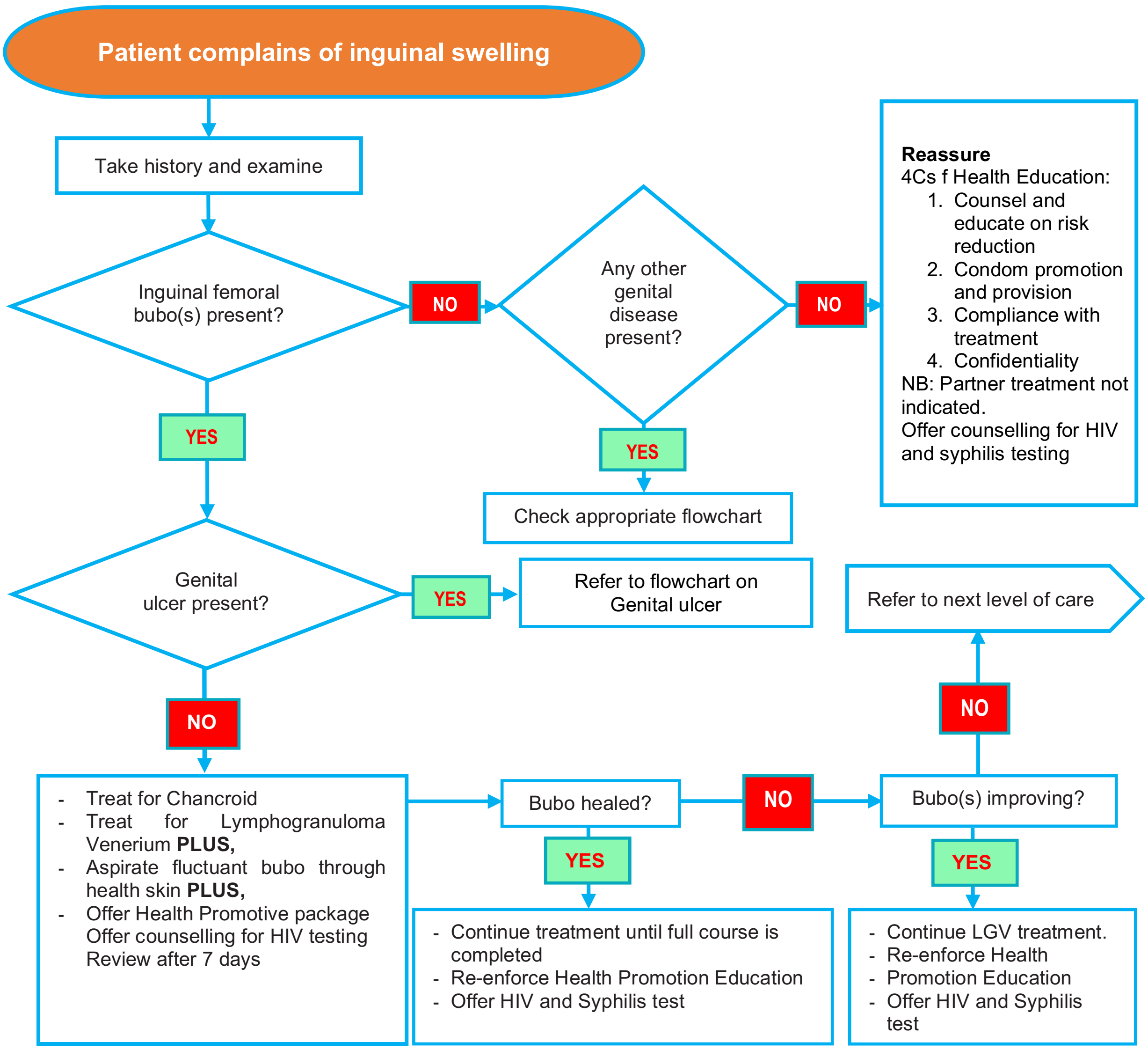
Female Lower Abdominal Pains



TREATMENT for PID	
Drug	**Condition**
Ceftriaxone 500mg IM stat	Gonorrhoea
Doxycycline 100mg PO BD 14 days	Chlamydia
Metronidazole 400mg PO BD 14 days	Anaerobes
HEALTH PROMOTION PACKAGE FOR ALL PATIENTS	
1) Educate, ensure compliance and counsel 2) Promote abstinence during the course of treatment 3) Promote and demonstrate condom use, and provide condoms 4) Stress the importance of partner treatment and issue one notification slip to each sexual partner follow up partner treatment during review visits	5) Provide HIV and syphilis testing, for negative test results repeat after 3 months 6) Offer male circumcision to uncircumcised partner/patient 7) Provide cervical cancer screening for female partner if indicated

Figure 11: Lower Abdominal Pains Syndrome Flowchart

INGUINAL BUBO FLOWCHART

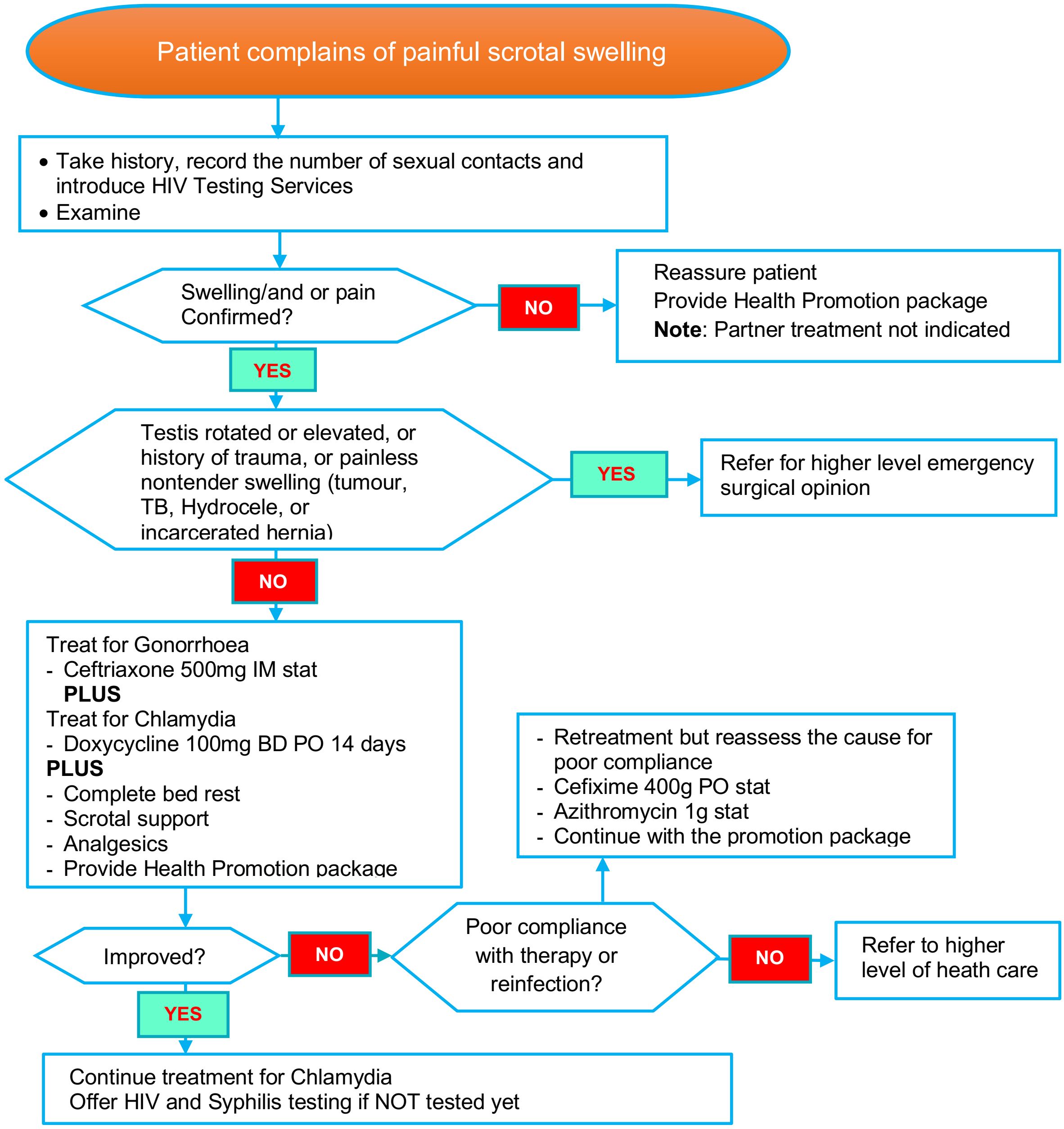


TREATMENT	
Drug	**Condition**
Doxycycline 100mg PO for 14 days	Lymphogranuloma Venerium
PLUS - Azithromycin 1g PO stat	Chancroid
PLUS - Aspirate fluctuant bubo through healthy skin	

HEALTH PROMOTION PACKAGE FOR ALL PATIENTS	
1) Educate, ensure compliance and counsel 2) Promote abstinence during the course of treatment 3) Promote and demonstrate condom use, and provide condoms 4) Stress the importance of partner treatment and issue one notification slip to each sexual partner, follow up partner treatment during review visits	5) Provide HIV and syphilis testing, for negative test results repeat after 3 months 6) Offer male circumcision to uncircumcised partner/ patient 7) Provide cervical cancer screening for female partner if indicated

Figure 12: Inguinal Bubo Flowchart

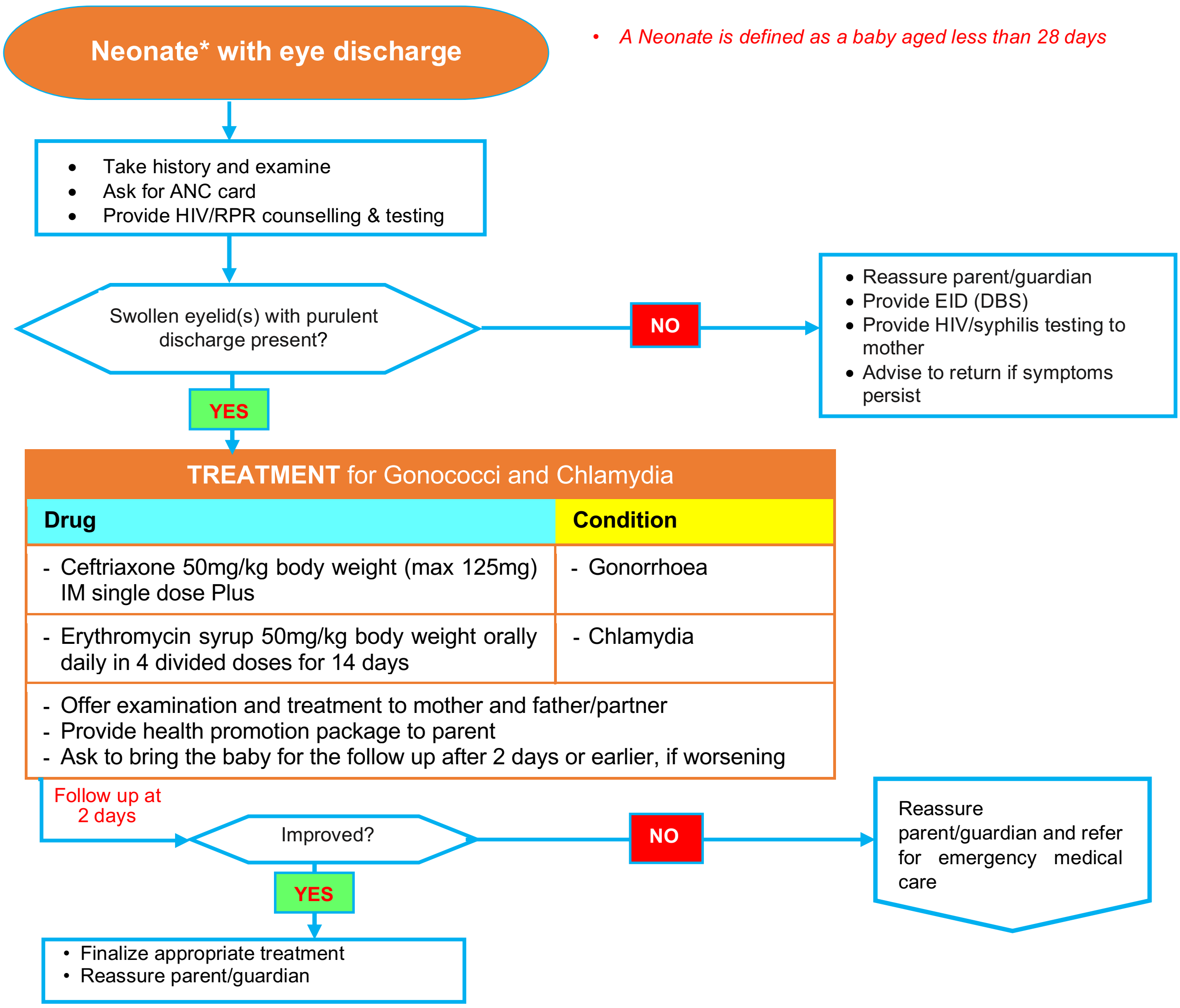
Scrotal Swelling Syndrome Flowchart



TREATMENT	
Drug	**Condition**
Ceftriaxone 500mg IM stat	Treat for Gonorrhoea
PLUS - Doxycycline 100mg BD PO 14 days	Treat for Chlamydia
PLUS - Complete bed rest, Scrotal support, Analgesics, Review after 7 days, compliant treatment	
HEALTH PROMOTION PACKAGE FOR ALL PATIENTS	
1) Educate, ensure compliance and counsel 2) Promote abstinence during the course of treatment 3) Promote and demonstrate condom use, and provide condoms 4) Stress the importance of partner treatment and issue one notification slip to each sexual partner, follow up partner treatment during review visits	5) Provide HIV and syphilis testing, for negative test results repeat after 3 months 6) Offer male circumcision to uncircumcised partner/patient 7) Provide cervical cancer screening for female partner if indicated

Figure 13: Scrotal swelling Syndrome Flowchart

NEONATAL CONJUNCTIVITIS SYNDROME FLOWCHART



HEALTH PROMOTION PACKAGE FOR PARENTS OF BABY WITH CONFIRMED NEONATAL CONJUNCTIVITIS	
1) Educate, ensure compliance and counsel 2) Promote abstinence during the course of treatment 3) Promote and demonstrate condom use, and provide condoms 4) Stress the importance of partner treatment and issue one notification slip to each sexual partner, follow up partner treatment during review visits	5) Provide HIV and syphilis testing, for negative test results repeat after 3 months 6) Offer male circumcision to uncircumcised partner/patient 7) Provide cervical cancer screening for mother

Figure 14: Neonatal Conjunctivitis Syndrome Flowchart

The Clinical and Aetiological Approaches to Managing Sexually Transmitted Infections

The Aetiological Approach

This approach requires the use of laboratory tests to identify the causative agent. It is the ideal approach in clinical medicine. It allows evidence-based diagnosis and management of STIs. It also enables correct documentation on the prevalence of various STIs. In addition, it allows the assessment of antimicrobial resistance or sensitivity patterns and therefore prevents the overuse/abuse of antimicrobials, thereby preventing the development of antimicrobial resistance. Furthermore, it can identify asymptomatic infections. However, it is only applicable in health care settings with suitable laboratory facilities and trained laboratory personnel.

The Clinical Approach

This approach involves the use of clinical experience to identify symptoms typical for a specific STI, then treatment is targeted to the suspected pathogen. It is usually the choice of resort when laboratory services are not available. It saves time and reduces lab expenses. However, it requires high clinical skills, may overlook mixed infections, and does not identify asymptomatic infections.

1.1.3 Ano-genital Herpes

Ano-genital herpes is a chronic, incurable viral infection. There are two types of Herpes Simplex Viruses (HSV)which are of clinical importance: HSV-1 and HSV-2. HSV infections are widespread among humans globally. The infection is lifelong and characterized by periodic reactivations at the infection site depending on the precipitating factors such as stress, immunosuppression, and trauma. HSV type 1 is primarily transmitted by oral-to-oral contact and commonly causes orolabial herpes (cold sores). HSV-1 also causes rarer conditions, such as keratitis and other ocular sequalae, and encephalitis. HSV-1 infection from oral-to-genital contact is becoming increasingly common, although reactivations are less frequent than for HSV-2. HSV-2 is almost entirely sexually transmitted, causing ano-genital herpes and sacral herpes. Ano-genital HSV infection may be unrecognized or result in painful ano-genital ulcer disease in a proportion of those infected. Asymptomatic viral shedding and transmission is common.

g) Relationship between HSV-2 and HIV infection

Evidence suggests that HSV-2 infection increases the risk of acquiring Human Immunodeficiency Virus (HIV). Symptomatic and asymptomatic viral shedding are common for both HSV type 1 and type 2. Clinical manifestations and patterns of anogenital ulcer disease may also be further altered in the presence of HIV infection.

h) Clinical presentation

Acute Episode of Ano-genital Herpes

The first episode usually occurs in the first 3–4 days after an incubation period within 5–14 days of sexual contact.

- Systemic and local symptoms of fever
- Headache, malaise, and myalgia
- Locally, there may be pain, itching, dysuria
- Vaginal or urethral discharge
- Tender inguinal lymphadenopathy
- Blistering or ulcerative lesions on the external genitalia. The lesions start as papules (pimples) or shallow vesicles (blisters), which spread rapidly over the anogenital area. The lesions may last up to 15–20 days until crusting and/or healing (Crusting does not occur on mucosal surfaces)

Recurrent HSV-2 Infections

Recurrent ano-genital herpes features:

- More localized symptoms of itching
- Multiple small vesicular lesions may c*oalesce* into large ulcers
- Recurrent ulcers – a cluster of vesical-pustular or ulcerative lesions on the external genitalia (penis, urethral meatus, scrotum, pubic area, and vulva) or on the anal and perianal areas
- In immunosuppressed individuals, these ulcers may persist and continue to expand laterally and superficially for a considerable period of time if not treated
- Since the ulceration of herpes is shallow (intra-epidermal), residual scarring from these lesions is uncommon
- Local mild pain
- Duration of the episode averages between four and five days but may be as long as up to 12–15 days

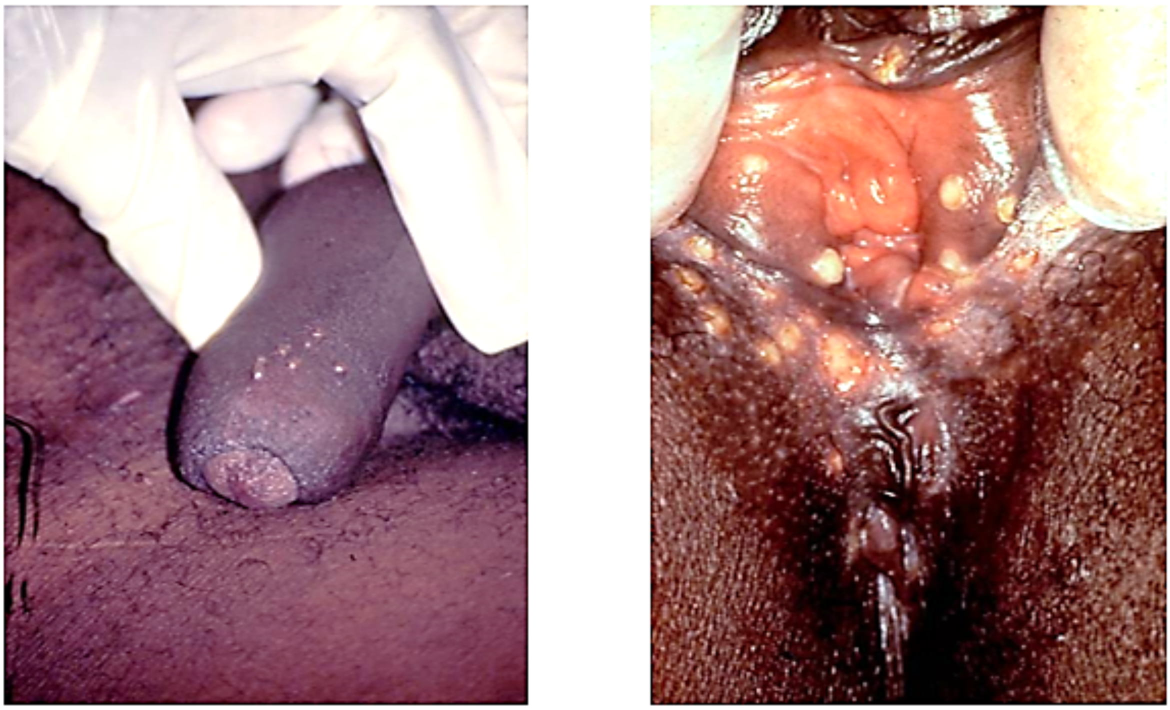


Figure 15: Vesicles and Ulcerative lesions on male (A) and female (B) patients respectively

i) Diagnosis

The clinical diagnosis of ano-genital herpes can be difficult, because the painful multiple vesicular or ulcerative lesions typically associated with HSV are absent in many infected persons. Recurrences and subclinical shedding are much more frequent for genital HSV-2 infection than for genital HSV-1 infection. Persons with ano-genital herpes should be tested for HIV infection

Serology

- Antigen tests: Rapid antigen tests for HSV-1 and HSV-2
- Antibody tests: The immunoglobulin G (IgG) and IgM-based type-specific antibody tests for HSV. IgG can be used to detect chronic infection and IgM is used as a marker for acute infection
- Culture methods: Viral culture enables identification and microbial susceptibility testing**.** Molecular testing: amplified molecular detection by PCR of HSV-1 and 2 DNA from the swabs of genital lesions is the most sensitive and specific test. Using a combined HSV and *T. pallidum* PCR, when available, would be of added benefit to implicate or exclude syphilis at the same time
- Molecular testing: amplified molecular detection by PCR of HSV-1 and 2 DNA from the swabs of genital lesions is the most sensitive and specific test. Using a combined

HSV and *T. pallidum* PCR, when available, would be of added benefit to implicate or exclude syphilis at the same time

j) Management of Anogenital Herpes

Management of anogenital HSV should address the chronic nature of the disease rather than focusing solely on treatment of acute episodes or genital lesions.

1. Counselling (regarding chronicity of disease and continued care)
2. Antiviral chemotherapy offers clinical benefit and is the mainstay of treatment (see table below for various stages of disease)

k) Treatment of Anogenital Herpes

Table 5: Recommendations for Systemic Antiviral Treatment of Ano-genital Herpes

	Drug	Dosage	In pregnancy
First clinical episode	Acyclovir	400mg PO TDS **OR** 200mg PO 5 times per day for 7 to 10 days	Acyclovir 400mg PO TDS for 10 days **OR** Acyclovir 200mg PO 5 times a day for 10 days
Recurrent episode (Start at prodromal stage)	Acyclovir	400mg PO TDS **OR** 800mg PO BD for 5 days **OR** 800mg PO TDS for 2 days	Acyclovir 400mg PO TDS for 5 days **OR** Acyclovir 800mg PO BD for 5 days **OR** Acyclovir 800mg PO TDS for 2 days
Chronic suppressive therapy (Recurrent ano-genital herpes)	Acyclovir	400mg PO BD (≥6 outbreaks per year) for upto 12 months	Acyclovir 400mg PO BD for upto 12 months

Severe disease	Acyclovir	5-10mg/kg IV TDS for 7-10 days **HSV encephalitis 21 days treatment**	
Anti-viral resistant HSV	Foscarnet Cidofovir	40-80mg/kg IV TDS for 14-21 days 5mg/kg IV once weekly	
			Note: ***Use Acyclovir only when the benefit outweighs the risk***

Source: *2020 Zambia National Essential Drug List*

NB: HIV-infected individuals may require longer duration of treatment of 5- 14 days.

Ano-Genital Herpes in Pregnancy

The risk for transmission to the neonate from an infected mother is high (30%–50%). Prevention of neonatal herpes depends both on preventing acquisition of anogenital HSV infection during late pregnancy and avoiding exposure of the neonate to herpetic lesions and viral shedding during delivery.

1.1.4 Chancroid

Chancroid is a highly contagious, ulcerative STI caused by a fastidious gram-negative bacillus *Haemophilus ducreyi*. It has an incubation period of about 1-5 days. It is more common in uncircumcised men. Like ano-genital herpes and syphilis, chancroid is risk factor for transmission of HIV.

In HIV infection, the ulcers may be less purulent and resemble syphilitic chancres. Also, people with immunosuppression may have rapidly aggressive and erosive ulcers of chancroid to the point of anatomically destroying the genital organs.

a) Clinical picture

Figure 16: An ulcer of Chancroid with an Inguinal Bubo

b) Clinical features

- Erythematous papule develops within hours to days of sexual exposure
- The papule evolves into a pustule that breaks down and becomes a painful ulcer
- Among men, the ulcers are usually on the penis (foreskin, shaft and sometimes on the glans), and as many as 50% develop unilateral or bilateral painful inguinal lymph nodes
- Large, painful, fluctuant lymph nodes (buboes) may also occur
- If not treated, buboes may suppurate and form fistulae or ulcers
- In women, ulcers may present on the vulva, and anal ulcers from autoinoculation may also occur
- Ulcers among women may be asymptomatic, especially when they are internal
- Women do not frequently present with inguinal adenopathy because of the different lymphatic drainage
- Painful, dirty looking and easily bleeding genital ulcers (soft chancre compared to painless indurated chancre of primary syphilis)
- Tender, suppurative inguinal adenopathy which is usually present with genital ulcers
- Kissing lesions may be present on opposing surfaces

c) Diagnosis

Clinical evaluation

Clinicians must have a high index of suspicion when they see an unusually painful, suppurative ulcer among men or women.

- A clinically compatible illness in a patient who is a contact of a confirmed case
- Typical 'shoal of fish' pattern on gram stain of a clinical specimen, where syphilis, Granuloma inguinale (GI) and anogenital herpes have been excluded, or
- A case definition of chancroid is a confirmed isolation of *H. ducreyi* from a clinical specimen
- A definitive diagnosis of chancroid requires identification of *H. ducreyi* on selective culture media and multiplex Polymerase Chain Reaction (M-PCR)

d) Management

Table 6: Management of Chancroid

Preferred drug	Alternative drug
Azithromycin 1g PO stat **OR**	Ceftriaxone 500mg IM stat **OR**
Cefixime 400mg PO stat	Erythromycin 500mg PO QID for 7 days

HIV-infected individuals and other immunocompromised individuals may require repeated or longer courses of treatment. Needle aspiration should be performed on fluctuant inguinal buboes through health skin. The aspirated pus can be sent for culture.

1.1.5 SYPHILIS

Syphilis is a systemic disease that can be contracted during sexual intercourse but can also be transmitted from mother to child during pregnancy, through blood transfusion and from contact with contents of a syphilitic lesion. It is caused by the spirochete bacterium *Treponema pallidum* which can be detected directly and or using Treponemal and non-Treponemal serological tests. Incubation period is about 3 weeks.

a) Clinical features

Primary Syphilis

This is characterized by the presence of an ulcer or chancre at the site of inoculation. A chancre is usually clean, indurated, and painless. It forms approximately 21 days after initial exposure to *T. pallidum*.

Figure 17: Syphilis lesion

Secondary Syphilis

Manifestations include but are not limited to a typically non-itchy generalized macular or maculopapular rash mostly affecting the trunk but may also involve the palms and soles (Rosy copper penny sign), lymphadenopathy, and oral and genital mucous exophytic lesions (Condylomata lata) (2), loss of hair (Moth-eaten Alopecia) (3). Patients often give a history of a chancre or may have a persisting chancre.

Figure 18: Syphilis lesion Rosey Copper penny sign of Secondary Syphilis

(Picture courtesy of Malumani Malan)

Figure 19: Condylomata Lata lesions in a male adult

(Picture courtesy of Malumani Malan)

Early Latent Syphilis

Characterized by positive serological tests for syphilis, with a history of symptoms suggestive of infection or a positive syphilis contact within the preceding year.

Late Syphilis

Figure 20: Moth-eaten alopecia in Secondary Syphilis

(Picture courtesy of Malumani Malan)

Late latent syphilis or syphilis of unknown duration: Positive serological test for syphilis with no history of symptoms or contacts in the past one year

Tertiary Syphilis

Characterized by Gumma (granuloma resulting from endarteritis) which may affect the skin and bones. Cardiovascular syphilis may manifest as aortitis and aortic aneurysm. Neuro-syphilis may occur 10 to 30 years after initial infection and may include tabes dorsalis and general paralysis of the insane. However, in the HIV/AIDS population the disease progression may be expedited.

Congenital Syphilis

Congenital syphilis is syphilis present in utero and at birth. It occurs when *T. pallidum* is transmitted trans-placentally or at birth.

Clinical features

- **Early (0 to 2 years old):** Newborns may be asymptomatic and only identified on routine neonatal screening
- **Late (≥ 2 years old):** Some of the commonest clinical features include Hutchinson's teeth, keratitis, frontal bossing, deafness, saddle nose, swollen knees, and saber shin

b) Diagnosis of Syphilis

The table below shows point of care tests:

Table 7: Point of Care tests for Syphilis

Treponemal antigen tests (Specific to Syphilis)		
Sn	**Test**	**Application**
1.	TPHA or TPPA	Can be used both as standard screening and confirmatory tests. More practical than the FTA-Abs
2.	Enzyme immunoassay (EIA)	Sensitive and specific screening test Positive from primary infection onward
3.	IgM-Enzyme Immunoassay	Improves detection of early primary infection
4.	FTA-abs	Standard confirmatory test
Cardiolipin antigen tests (Non-specific to Syphilis)		
1	VDRL	Best used for CSF specimen
2	RPR	Widely used screening test, also used to measure titers

1.1.6 Lymphogranuloma Venereum

Lymphogranuloma Venereum (*LGV*) is an infection caused by *Chlamydia trachomatis* (serovars L1, L2, L3). The infection has an incubation period of about 3-21 days. It has a worldwide distribution but is more common in tropical and sub-Saharan countries.

a) Clinical features

- Primary lesion is a painless ulcer or sore that develops at the site of inoculation (male or female genital tract) which is self-limited and is commonly unnoticed
- Weeks later, tender inguinal lymphadenopathy develops, with abscess formation (buboes). The buboes may be unilateral or bilateral
- Swelling and redness of labia
- Herpetic like vesicles or small pigmented scars

Figure 21: Bilateral Buboes in a female patient

b) Diagnosis

Diagnosis is based on the clinical suspicion of LGV, but the following tests can be performed:

- Rapid antigen test
- Specific serological tests
- Nucleic acid amplification tests (NAATs)

c) Treatment

Preferred Regimen	Alternative regimen
Doxycycline 100mg PO BD for 21 days	Erythromycin 500mg PO QID for 21 days for pregnant and lactating women
	Azithromycin 1g PO weekly for 3 weeks (3 total doses)

NOTE

- Sexual contacts within 60 days before onset of symptoms should be examined, tested, and presumptively treated with a chlamydia regimen
- Prolonged therapy might be required for patients with both HIV and LGV
- Pregnant and lactating women should be treated with Erythromycin
- Manage fluctuant buboes may be drained through healthy skin to relieve symptoms

d) Complications

- Buboes shrink and form fibrous masses, rupture and form sinuses or fistulae leading to healing by fibrosis and chronic inflammation of lymph nodes
- This may cause lymphatic obstruction leading to genital elephantiasis
- Rectal infection can cause swelling and scarring resulting in the risk of long-term bowel complications like complete bowel obstruction from rectal stricture
- In women, cervicitis, peri-metritis, salpingitis, lymph stasis and hypertrophic ulceration of the external genitalia may occur
- In men, lymphatic obstruction causing oedema of the penis and scrotum; others include rectal urethra stricture, fistula, and rupture of buboes
- Unilateral chlamydial conjunctivitis (babies may acquire infection at birth)
- Systemic spread may occur and possibly result in arthritis, pneumonitis, hepatitis, or peri-hepatitis

1.1.7 Granuloma Inguinale (Donovanosis)

a) Background

Granuloma inguinale is a rare, progressive ulcerative STI of the genital and perineal skin caused by the intracellular gram negative bacteria *Klebsiella* (formerly *Calymmatobacterium) granulomatis*.

b) Clinical features

After an icubation period of upto 12 weeks, a painless, red skin nodule slowly enlarges becoming a raised, beefy red, moist, smooth and foul smelling lesion which easily bleeds when touched. The lesion slowly enlarges, often ulcerates, and may spread to other skin areas without regional lymphadenopathy.

The common sites of infection are:

- Penis, scrotum, groin and thighs in men
- Vulva, vagina and perineum in women
- Anus and buttocks
- Face in both sexes

Figure 22: Painless Ulcerative lesions on the shaft and glans

The lesions heal slowly, with scarring. Secondary infections with other bacteria are common and can cause extensive tissue destruction.

c) Diagnosis

The diagnosis of this disease is usually based on clinical appearance. The causative organism is difficult to culture, and requires visualization of dark staining Donovan bodies on crushed tissue preparation or biopsy.

d) Management

Table 8: Management of Granuloma Inguinale

Preferred drugs	Alternative drugs
Doxycline 100mg PO BD for 21 days or until all lesions completely healed **OR** Erythromycin 500mg PO OD for 21 days or until all lesions completely healed	Azithromycin 1g PO weekly for 3 weeks (3 total doses) **OR** Erythromycin 500mg PO QID for 21 days or until all lesions completely healed **OR** Trimethoprim-sulfamethoxazole (160/800mg) PO OD for 21 days or until all lesions completely healed

Source : *CDC guidelines 2021*

Persons with Granuloma inguinale and have HIV infection should receive the same regimens as those who do not have HIV.

e) Complications

- Permanent hypopigmentation on genital skin
- Genital scarring
- Genital swelling

1.1.8 GONORRHOEA

Gonorrhoea is an STI caused by *Neisseria gonorrhoeae* which is exclusively a human pathogen and almost exclusively sexually transmitted. It causes urethral discharge in men and cervical discharge in women.

a) Clinical features

Cases should be classified as:

- Uncomplicated, when there is infection of one or both of the following:
    a. Urogenital tract
    b. Anorectal area (proctitis)

- PID (pelvic inflammatory disease) or epididymitis
- Extra-genital infection of one or both of the following:
    a. Pharynx
    b. Other sites including eyes in newly born (commonly ophthalmia neonatorum)

In men:

- Median incubation period is about 3-6 days. Incubation period of Gonorrhoea is usually shorter than that of non-gonococcal urethritis (NGU) caused by chlamydia
- Purulent or mucopurulent urethral discharge in 80% of cases
- Dysuria
- Rectal infection is usually asymptomatic may cause anal discharge or pain in around 10% of cases
- Pharyngitis may be present

Figure 23: Purulent Urethal Discharge

In Women:

- Up to 50% of all women with gonorrhoea are asymptomatic
- Incubation period in women is longer than in men
- Symptoms are non-specific and cannot be distinguished from those of other lower genital tract infections
- Urethritis, purulent urethral discharge, and/or dysuria may or may not be present
- The most common symptom is a change in vaginal discharge, originating from an infected endocervix
- Gonococcal proctitis and pharyngitis may also be seen in women

Two major diagnostic signs of cervicitis include:

a) Purulent or mucopurulent endo-cervical exudates visible in the endo-cervical canal or on an endo-cervical swab specimen and
b) Sustained endo-cervical bleeding easily induced by gentle passage of a cotton swab through the cervical Os

b) Diagnosis

A case definition for gonorrhoea is confirmed laboratory isolation of *N. gonorrhoea* from a clinical specimen.

- Point of Care Tests as outlined in the table for diagnosis of the Neisseria Gonorrhoeae antigen test (RDT). A high vaginal swab for females or a urethral swab for males is sufficient to be used in testing
- Microscopy: Gram-negative intracellular diplococci seen on examination on direct microscopy in Gram-stained samples of discharge collected from the urethra or rectum or endocervix from women
- Culture: Currently regarded as the standard method for diagnosis of infection and allows the testing for antibiotic susceptibility and detection of resistance patterns
- Nucleic acid amplification tests (NAATs): NAATs have very high sensitivity and specificity compared to microscopy and culture

c) Management

Table 9: Management of Urethral Discharge

Preferred drugs	Alternative drugs
Ceftriaxone 500mg IM stat Ceftriaxone 25-50mg/kg IM or IV stat, not to exceed 125mg (to treat ophthalmia neonatorum) Frequent eye washes with running water (saline lavage) with application of antibiotic eye drops like Neomycin/Gentamycin drops	Cefixime 400mg PO stat **OR** Cefotaxime 500mg IM OD for 5-7 day **Note**: The preferred is Ceftriaxone or Cefixime but if not available, Gentamicin 240mg IM stat **PLUS** Azithromycin 2g PO stat

Note: All women with history of contact with symptomatic partners should be treated.

d) Complications

Adults

In females: Pelvic inflammatory disease, ectopic pregnancies, infertility, chronic lower abdominal pain, miscarriages, disseminated gonococcal infection.

In males: Prostatitis, urethral scarring and stricture, infertility, epididymo-orchitis, disseminated gonococcal infection.

Purulent Conjunctival gonococcal infection in both men and women.

1.1.9 Neonatal Conjunctivitis

Known as Ophthalmia Neonatorum typically presents in the first four weeks of life (the first 28 days of life). The infection is usually acquired during delivery and is usually due to maternal STI. The most common causative agents are in the table below:

Age of Neonate	Likely causative agent
24 - 48 hours of life	*Neisseria gonorrhoeae*
5 - 14 days of life	*Chlamydia trachomatis*
6 - 14 days of life	*Herpes simplex virus*

a) Clinical features

- Usually presents as unilateral or bilateral eye swelling
- Reddening of the eyes and eye lids

The table below summarizes the clinical feature by causative agent.

Type of Discharge	Likely cause
Profuse purulent, Bilateral eye affected	Gonorrhoea
Water followed by purulent then bloody, Unilateral eye then bilateral after 2 to 7 days	Chlamydia
Watery, unilateral eye affected	Herpes

Figure 24: Eyes of baby affected with Gonorrhoea

b) Diagnosis

- Microscopy, Culture and Sensitivity for the discharge (Oropharynx swab)
- Nucleic Acid Amplification Tests for Chlamydia

c) Management

Table 10: Management of Neonatal Conjunctivitis

Preferred drugs	Condition
Ceftriaxone 25-50mg/kg stat or Cefotaxime 100mg/kg IM/IV stat	Gonococcal conjunctivitis
Erythromycin 50mg/kg/day in divided doses PO QID for 14 days	Chlamydia conjunctivitis
Acyclovir 60mg/kg/day in divided doses PO QID for 14 to 21 days	Herpetic keratoconjunctivitis

d) Supportive treatment:

- Isolate neonate for the first 24 hours of parenteral treatment
- Test for HIV (PCR) and syphilis
- Ophthalmology and paediatric infectious disease consultation

- Evaluate for systemic infection (Sepsis, Meningitis, arthritis)
- Irrigate eyes with Normal Saline at 1-to-2-hour intervals, use neomycin/gentamycin eye drops
- Avoid Eye Patching
- Avoid cross contamination by frequent hand washing and wearing gloves
- Offer counselling and treatment (targeting the 3 causative organisms) to biological parents

e) Complications

- Corneal perforation
- Corneal scarring
- Bacterial superinfection
- Chronic infection
- Persistent Keratoconjunctivitis
- Symblepharon
- Astigmatism
- Blindness

1.1.10 Chlamydia

Chlamydia is a sexually transmitted infectious disease caused by the bacterium *Chlamydia trachomatis*. Globally, it is the most common STI and also causes an ocular infection called "Trachoma" which is the leading infectious cause of acquired blindness in the world. The bacteria are transmitted through direct contact with infected tissue such as vaginal secretions, seminal fluids and can even be transmitted from an infected mother to her new-born baby. *C. trachomatis* is differentiated into 18 Serovars which are associated with different medical conditions.

Serovar	Medical Condition
Serovars A, B, Ba, C	Trachoma
Serovars D to K	Genital Tract Infections
Serovars L1 to L3	Lymphogranuloma Venereum

a) Clinical Presentation

The following are the conditions caused by Chlamydia:

In males

- Urethritis: dysuria, urethral discharge, urinary frequency, pyuria, urgency
- Epididymitis: testicular pain and tenderness, palpable swelling of the epididymis and fever
- Prostatitis: dysuria, pelvic pain, painful ejaculation
- Reactive arthritis (formerly known as Reiter's Syndrome- an autoimmune condition): triad of arthritis, urethritis, and uveitis (joint pain, urethritis features and eye symptoms)

In females

- Cervicitis: mostly asymptomatic to mild symptoms, vaginal discharge or bleeding, dysuria, mucopurulent cervical discharge, post-coital or intermenstrual bleeding
- PID: lower abdominal pain, pelvic pain with or without signs of cervicitis which may complicate to Infertility and Ectopic pregnancies

In both sexes

- Proctitis: rectal pain, rectal discharge, and rectal bleeding
- Conjunctivitis: non-purulent eye discharge
- Pharyngitis: throat irritation
- Lymphogranuloma Venereum: painless genital ulcers, small stellate in appearance with inguinal lymphadenopathy
- Pneumonia: nasal congestion, cough, apnoeic spells. In Infants, Rales on auscultation maybe heard
- Perihepatitis: right upper quadrant pain or pleuritis in the presence of PID
- Trachoma

b) Investigations

- Urinalysis and urine microscopy
- Antigen tests (endocervical, urethral, rectal, or oropharyngeal swabs)
- Cytology and cultures (endocervical, urethral, rectal, or oropharyngeal swabs)
- Molecular detecting tests (DNA, RNA)
- Pregnancy test in women suspected to have Chlamydia is MANDATORY for early diagnosis and treatment

c) Management

Table 11: Management of Chlamydia

Condition	Preferred drugs	Alternative drugs
1. Uncomplicated Genital Chlamydia	Doxycycline 100mg PO BD for 7 days	**OR** Azithromycin 1g PO stat **OR** Erythromycin 500mg PO QID for 7 days
2. Anorectal Chlamydia Infection	Doxycycline 100mg PO BD for 7 days	
3. Chlamydia infection in pregnancy	Azithromycin 1g PO stat	Erythromycin 500mg PO QID for 7 days
4. Pneumonia (in infants)	Erythromycin 50mg/kg in divided doses PO QID for 14 days	Azithromycin 20mg/kg/day PO OD for 3 days

1.1.11 Non-Gonococcal Urethritis

Non-Gonococcal Urethritis is a nonspecific diagnosis that can have various infectious aetiologies. *C. trachomatis* has been well established as an NGU aetiology; however, prevalence varies across populations and accounts for <50% of overall cases. *M. genitalium* is estimated to account for 10%–25% of cases and *T. vaginalis* for 1%–8% of cases depending on population and location. Other organisms implicated in NGU include among others *Ureaplasma urealyticum, viruses and enteric bacteria.* However, even when extensive testing is performed, no pathogens are identified in approximately half of cases.

a) Clinical features

NGU may be asymptomatic. However, when present, the following may be complained about:

- A clear, creamy white, yellow, or green discharge with or without dysuria
- Urethral itch or discomfort may be the only symptom
- Reactive arthritis, which is characterized by urethritis, uveitis, and arthritis may also be experienced

b) Diagnosis

NGU is confirmed for symptomatic men when diagnostic evaluation of urethral secretions indicates inflammation, without evidence of diplococci by Gram, *mycobacterium* (MB), or GV smear on microscopy

M. genitalium testing should be performed for men who have persistent or recurrent symptoms after initial empiric treatment. Testing for *T. vaginalis* should be considered in areas or among populations with high prevalence, in cases where a partner is known to be infected, or for men who have persistent or recurrent symptoms after initial empiric treatment.

1. Culture of the discharge
2. Use of Multiplex-PCR

c) Management

Azithromycin and Doxycycline are both highly effective for the treatment for Chlamydial urethritis.

Table 12: Management of Non-Gonococcal Urethritis

Preferred drugs	**Alternative drugs**
Doxycycline 100mg PO BD for 7 days	Azithromycin 1g PO stat **OR** Azithromycin 500mg PO stat; then 250mg PO OD for 4 days

1.1.12 Trichomoniasis

Trichomoniasis is caused by the protozoan parasite *Trichomonas vaginalis*. The most common route of transmission is sexual contact. It is associated with a two to three-fold increased risk of HIV acquisition and significantly increased risk of preterm birth and other adverse pregnancy outcomes, and PID among women with HIV infection.

a) Clinical features

When present, signs and symptoms may include:

- A diffuse white or clear typical frothy, malodorous, or yellowish green discharge
- Pruritus (severe itching) or with vulvar irritation

- Urethral irritation in men
- In women, both the vagina and cervix may be infected and infection may ascend to cause PID
- Infected men may have symptoms of urethritis, epididymitis, or prostatitis. However, most infected persons have minimal, or no symptoms and untreated infections may last for months

b) Diagnosis

- Wet-mount microscopy of genital secretions (should be done within a few minutes of collection to improve sensitivity) where the motile protozoan and its flagellate is seen under the microscope
- Trichomonas rapid antigen-detection tests
- Culture on specialized culture media (swabs and urine specimen)
- Nucleic acid amplification tests (swabs and urine specimen)

  Note: if wet mount is negative; do NAAT tests due to low sensitivity of the wet mount

c) Treatment

Preferred drugs	Alternative drugs
For men: **Metronidazole 2g PO stat**	Tinidazole 2g PO stat for both men and women
For women: **Metronidazole 400mg PO BD for 7 days**	

Note: *Avoid concomitant use of alcohol when on Metronidazole medication*

Follow up after 5 days

Retest is advised within 3 months but if not possible, whenever the patient presents to the hospital within 12 months

d) Complications

In women:

- Pelvic Inflammatory Disease (PID)
- Infertility
- Pregnant women may have adverse pregnancy outcomes, particularly premature rupture of membranes, preterm delivery, and delivery of infants who are small for gestational age

In men:

- Epididymo-orchitis

- Prostatitis
- Sterility

1.1.13 Bacterial Vaginosis

Bacterial vaginosis is the most common cause of vaginal discharge among women of childbearing age. It is a poly-microbial disorder of the vaginal microbiome. The condition is characterized by low concentrations or an absence of lactobacilli and a florid presence of anaerobic bacteria. Bacterial vaginosis is not a sexually transmitted condition, but it has been linked to several adverse outcomes, such as pregnancy associated complications and an increased risk of STIs, HIV, PID and infertility.

a) Clinical features

- On external visual examination and digital examination of the vagina, the thin, white, homogenous discharge may be observed externally on the posterior fourchette of the vulva or the labia
- A smell of spoiled fish and can be detected in vaginal specimens

b) Diagnosis of BV

The two gold standard methods used to diagnose BV are:

1. Amsel's clinical criteria and
2. Nugent Gram staining evaluation (laboratory -based)

Amsel's clinical criteria

The diagnosis is positive for BV if at least three out of the four criteria are fulfilled. These criteria are

- Presence of a typical discharge – amine odour can be sensed spontaneously
- Vaginal pH > 4.5
- A positive whiff test (a drop of 10% Potassium hydroxide to vaginal fluid on a slide)
- Microscopy: the presence of clue cells in the wet smear (Clue cells are epithelial cells of the vagina whose borders are difficult to see because so many bacteria are found on the surface of the cells)

Nugget's Criteria

Gram staining

Analyses 3 types of bacteria: Lactobacillus, Bacteroides, Gardnerella and Mobiluncus. The total scores are then graded as follows:

- 0-3 normal
- 4-6 intermediate bacterial count
- 7-10 bacterial vaginosis

c) Treatment of BV

Preferred drugs	Alternative drugs
Metronidazole 1g PO stat **OR** Metronidazole 400mg PO TDS for 7 days **OR** Clindamycin cream 2% one full applicator (5g) intravaginally OD Nocte for 7days	Clindamycin 300mg PO BD for 7 days **OR** Tinidazole 2g PO BD for 2 days **OR** Tinidazole 1g PO OD for 5 days

d) Prevention

Advise clients to avoid douching or using bubbble bath or any other over-the-counter vaginal hygiene products, liquid soaps and body washes.

Infections characterized by Growths on Genitals

Infections that cause growths or lumps on the ano-genital skin are mostly viral and include Molluscum Contagiosum virus and human papilloma virus.

1.1.14 MOLLUSCUM CONTAGIOSUM

Molluscum Contagiosum (MC) is a self-limiting infectious dermatosis, frequent in the paediatric population, sexually active adults, and immunocompromised individuals.
It is caused by *Molluscum contagiosum virus* (MCV) which is a virus of the Poxviridae family. MCV is transmitted mainly by direct contact with infected skin, which can be sexual, non-sexual, or autoinoculation. Close physical contact accounts for transmission in children, in whom lesions appear on the face, arms, and trunk.

In adults, transmission is mainly through sexual contact enhanced by friction and micro trauma. Lesions appear on the genitals and surrounding skin. It is rarely a serious clinical problem except in patients with HIV infection, where in advanced immunosuppression may be extensive, intractable, and become secondarily infected.

a) Clinical features

Typical eruptions and appear as dome shaped firm rounded papules with sizes from 2 to 5 mm, pink or skin-coloured, with a shiny central umbilicated surface from which caseous material can be expressed.

Figure 25: Picture of Molluscum Contagiosum (A) in a Caucasian (B) in a person of colour
(Picture courtesy of Malumani Malan)

In children, the main affected areas are sites of exposed skin, such as the trunk, extremities, intertriginous regions, genitals, and face, except the palms and soles. The involvement of the oral mucosa is rare. In children, genital lesions are mainly due to autoinoculation and are not pathognomonic of sexual abuse.

In adults, lesions are most frequently located in the lower abdomen, thighs, genitals, and perianal area. Generalised lesions, especially on the face may indicate immune-suppression due to underlying HIV infection.

b) Diagnosis

- Clinical diagnosis is adequate
- Histological examination of a biopsied lesion should only be done if the clinical diagnosis is in doubt. A mass of whitish macerated material (a core) can be extracted with the tip of sterile needle and small forceps from centre of papule

  Note: The clinician should avoid direct contact with caseous material as this is highly infectious.
- Electron microscopy (if available) of the core material can be done to demonstrate pox virus

c) Management

Excisional Curettage

- Destruction of the individual lesions by extraction of the umbilicated lesions (core) with needle or forceps, curettage, cryotherapy with liquid nitrogen or Trichloroacetic acid is effective. Each lesion should be thoroughly opened with a fine sterile needle. The contents should be expressed, and the inner wall touched with either Phenol/silver nitrate 30% Trichloroacetic acid
- Cryotherapy with liquid nitrogen

Topical Therapy

- Podophyllotoxin 0.5% solution or 0.15% cream to the individual lesion BD for 3 days (may be useful as patient-applied therapy (only for men). Not indicated in pregnant women as it is foetal toxic
- 5% Imiquimod cream may also be useful in treating larger lesions of Molluscum Contagiosum patients (not recommended in children)

Oral Therapy

Gradual removal of lesions may be achieved by oral therapy with Cimetidine. This is mainly recommended for children who cannot tolerate physical therapies.

1. Molluscum contagiosum therapy for Immunocompromised Persons: Refer to the latest Zambia Consolidated Guidelines for Treatment and Prevention of HIV.

1.1.15 Ano-genital Warts (Condylomata Acuminata)

Anogenital warts are caused by *Human papilloma viruses* (HPV). 90% of anogenital warts are caused by low-risk non-oncogenic HPV-6 and 11. High risk HPV types 16, 18, 31, 33, and 35 are also occasionally found in anogenital warts (usually as co-infections with HPV-6 or 11) and can be associated with high-grade squamous intraepithelial lesions, particularly in persons who have HIV infection.

a) Clinical features

- Anogenital warts are usually asymptomatic, but depending on the size and anatomic location, they can be painful or pruritic
- Growths are usually flat, papular, or pedunculated growths on the genital skin and mucosa. They occur commonly at certain anatomic sites, including around the vaginal introitus, under the foreskin of the uncircumcised penis. Warts can also occur at multiple sites in the anogenital epithelium or within the anogenital tract (e.g., cervix, vagina, urethra, perineum, perianal skin, anus, and scrotum)
- Intra-anal warts are observed predominantly in persons who have had receptive anal intercourse but can also occur in men and women who do not have a history of anal sexual contact. Therefore, persons with external anal or perianal warts might benefit from inspection of the anal canal by digital examination or if available standard anoscopy.

Figure 26: Pictures of Ano-genital warts in a pregnant woman A) 25 weeks B) post caesarean section at 39 weeks

(Pictures courtesy of Jane Mwamba Mumba)

Figure 27: Genital warts in a pregnant woman

Figure 28: Genital warts (A) on a penile shaft (B) on the labia

(Pictures courtesy of Malumani Malan)

b) Diagnosis

- Diagnosis of ano-genital warts is usually made by visual inspection
- The diagnosis can be confirmed by biopsy, which is indicated if lesions are atypical and do not respond to therapy

c) Management

Recommended regimens for external genital warts can be classified into:

1. **Patient-applied regimens**
   - Imiquimod 3.75% or 5% cream: Imiquimod 5% cream should be applied once Nocte, three times a week for up to 16 weeks. Similarly, Imiquimod 3.75% cream should be applied once at bedtime, but is applied every night. The treatment area should be washed with soap and water 6-10 hours after the application). **Note: do not give children under the age of 12**
   - Podophyllin 0.5% solution or gel
2. **Provider-administered regimens**
   - Podophyllin 25% solution applied once weekly by health care provider. The area to which Podophyllin is administered should not contain any open lesions, wounds, or friable tissue, and the preparation should be thoroughly washed off 1-4 hours after application. Vaseline petroleum jelly could be applied around the wart to avoid the Podophyllin getting in contact with normal skin
   - Cryotherapy with liquid nitrogen or cryoprobe (**Note:** contraindicated in pregnancy and small children)
   - Surgical removal by excision, laser or electro surgery
   - Trichloroacetic acid (TCA) or Bichloroacetic acid (BCA) 80%-90% solution applied weekly
   - Electrodessication with curettage

d) Complications

High-risk HPV infections can increase the risk of the following:

- Ano-genital malignancies like cervical, anal, and vulva cancer in women, and penile or anal cancer in men
- In pregnant woman, warts can block the vaginal passage for the baby for which caesarean section is indicated
- If the baby aspirates the fluid containing the HPV, the wart may grow in baby's throat (oropharyngeal papillomatosis)

e) Prevention

- Ano-genital warts can be prevented through the use of HPV vaccination for girls (but does not treat existing genital warts)
- Male circumcision
- Screen for Cervical Cancer

SEXUALLY ACQUIRED VIRAL HEPATITIS AND HIV

Viral hepatitis may be caused by viruses that are transmitted through various sexual practices. Hepatitis B virus is the main sexually transmitted blood-borne virus.

1.1.16 HEPATITIS A

Hepatitis A has predominantly a faecal-oral route mode of transmission but due to different sexual practices this has been included in the STI guidelines.

Hepatitis A is caused by Hepatitis A virus (HAV). It has an incubation period of approximately 28 days (range: 15-50 days). HAV replicates in the liver and is shed in high concentrations in faeces from 2 to 3 weeks before to 1 week after the onset of clinical illness. HAV infection produces a self-limited illness that does not result in chronic infection or chronic liver disease. HAV infection is primarily transmitted by the faecal-oral route by either person to person contact or through consumption of contaminated food or water. Transmission of HAV during sexual activity probably results from fecal-oral contact (e.g., oral-anal, digital-anal). Antibody produced against HAV infection persists for life and confers protection against re-infection. Acute liver failure from acute Hepatitis A is rare (overall case fatality rate: 0.5%). The risk for symptomatic infection is directly related to age, with >70% of adults having symptoms compatible with acute viral hepatitis.

a) Clinical features

- Nausea and vomiting, change in smell or taste
- Malaise
- Right upper-quadrant pain
- Diarrhoea
- Itchiness due to cholestasis
- Jaundice

Most children have either asymptomatic or unrecognized infection.

b) Diagnosis

Diagnosis of Hepatitis A cannot be made on clinical grounds alone.

- Serologic tests: IgM antibody to HAV is diagnostic of acute HAV infection

c) Treatment

Patients with acute Hepatitis A usually require only supportive care, with no restrictions in diet or activity. Hospitalizations might be necessary for patients who become dehydrated because of nausea and vomiting and is critical for patients with signs or symptoms of acute liver failure. Medications that might cause liver damage or are metabolized by the liver should be used with caution in persons with Hepatitis A.

d) Prevention

- Hepatitis A vaccine (If within two weeks of exposure)
- Intravenous immunoglobulin (IVIg) 0.02mL/kg IM x 1 dose (If within two weeks of exposure)
- Observation of minimum hygiene standards

e) Complications

In extremely rare cases, Hepatitis A can lead to acute liver failure.

1.1.17 HEPATITIS B

Hepatitis B is caused by an infection with the Hepatitis B virus (HBV). The incubation period is 6 weeks to 6 months. The highest concentration of HBV is found in blood, with lower concentrations found in other body fluids including wound exudates, semen, vaginal secretions, and saliva [(25)]. HBV is more infectious and relatively more stable in the environment than other blood-borne pathogens like HCV and HIV.

The primary risk factors associated with infection among adolescents and adults are unprotected sex with an infected partner, multiple partners, history of other STIs, and injection-drug use.

a) Diagnosis

Diagnosis of acute or chronic HBV infection requires serologic testing. The following tests can be done:

- IgM anti-HBc: diagnostic of acute or recently acquired HBV infection
- Anti-HBs: produced after a resolved infection and is the only HBV antibody marker present after vaccination
- The presence of HBsAg and total anti-HBc, with a negative test for IgM anti-HBc, indicates chronic HBV infection
- The presence of anti-HBc alone might indicate a false-positive result or acute, resolved, or chronic infection

b) Treatment

Treatment of patients with both HBV infection and HIV is to prevent the progression of disease to cirrhosis, liver failure and hepatocellular carcinoma. HBV infection can be self-limiting or chronic. Therapy with antiretroviral drugs is currently recommended for patients with evidence of Chronic Active Hepatitis B infection. While in acute cases only supportive treatment is often required. (Refer to national guidelines on treatment of viral hepatitis and to the Zambia consolidated guidelines for treatment of HBV and HIV co-infection)

c) Prevention

Hepatitis B vaccine is available and confers a lifelong immunity against HBV infection. Strategies to eliminate HBV infection should include:

- Prevention of perinatal infection through routine screening of all pregnant women for HBsAg and immune-prophylaxis of infants born to HBsAg-positive mothers or mothers whose HBsAg status is unknown

Vaccinations:

- Routine infant vaccination
- Vaccination of previously unvaccinated children and adolescents through age 18 years
- Vaccination of previously unvaccinated adults at increased risk of infection

- Hepatitis B immunoglobulin (HBIg) provides temporary (i.e., 3-6 months) protection from HBV infection and is typically used as post-exposure prophylaxis either as an adjunct to Hiogepatitis B vaccination in previously unvaccinated persons or alone in persons who have not responded to vaccination. The recommended dose for HBIg is 0.06mL/kg

Social settings:

- Different razor blades to be used in social practices e.g., traditional circumcision ceremonies (Mukanda), tattoos (indembo, inembo, ndembo), body piercings
    - Safer sex practices
    - Discourage illegal drug use

Hospital settings:

- Isolate infected patient
- Hand hygiene: including surgical hand preparation, hand washing, and use of gloves
- Safe handling and disposal of sharps and waste
- Safe cleaning of equipment
- Testing of donated blood
- Improved access to safe blood
- Training of health personnel

d) Complications

- Liver cirrhosis
- Liver cancer
- Liver failure

Ectoparasite Infestations

Ectoparasitic infestation is a parasitic disease that live primarily on the surface of the host. Parasitic infestations such as pubic lice and scabies may also be transmitted during sexual contact. They may cause some serious outbreaks if not recognized early.

1.1.18 Pubic Lice

Pubic lice, commonly called crabs, are tiny insects found in the genital area. They are a different type of louse from head lice and body lice. Pediculosis pubis (pubic lice) is usually transmitted by sexual contact. It is caused by the blood-sucking louse Phthirus pubis which is usually transmitted by close body contact. It is different type from the head and body lice. Transmission through sharing beds or clothing is possible but unlikely.

a) Clinical features

Figure 29: Pubic lice

(Courtesy of e-medicine health/Charles Patrick Davies, MD, PhD)

- Patients seek medical attention because of pruritus. Excessive pruritis may cause wounds or an infection in the affected areas
- Lice or nits on the pubic area

Figure 30: Excoriated papules, raised burrows and occasional nodules on an oedematous penis

(Picture courtesy of Malumani Malan)

b) Diagnosis

- Lice can be identified by the naked eye, or if necessary, with a hand lens
- Patients should be offered tests for other STIs

c) Treatment

Preferred drugs
Malathion 0.5% lotion. Apply to affected parts. Let it dry naturally and rinse off after 12 hours
Permethrin 1% cream. Apply to clean damp hair. Rinse after 10 minutes and dry

d) Prevention

To prevent pubic lice infestation:

- Avoid sharing clothes, beddings or towels with anyone who has pubic lice
- Sexual contact should also be avoided until treatment is complete and successful

1.1.19 Scabies

Scabies is an infestation of the skin by the human itch mite known as *Sarcoptes scabiei*. The microscopic scabies mite burrows into the upper layer of the skin where it lives and lays its eggs. It is transmitted by close skin-to-skin contact. Scabies in adults is usually acquired through sexual contact while in children is through close physical contact. Symptoms are due to hypersensitivity to faeces deposited by the mite as it burrows through the stratum corneum.

Figure 31: Pictures of Scabies

(Pictures courtesy of Malumani Malan & Nyambe Sinyange)

a) **Clinical features**

- Severe, distracting itch, worse at night, occurring 2-6 weeks after acquiring infection
- Burrows caused by mites usually affect the wrists, finger webs, elbows, and genitals
- Excoriated papules, raised linear burrows, and occasionally nodules are seen
- Occasionally, secondary bacterial infection occurs, which complicates the clinical picture
- In those with immunosuppression or reduced skin sensation, very large numbers of mites are carried in crusted or Norwegian scabies

b) **Diagnosis**

- Clinical diagnosis depends on the recognition of the typical linear burrow ending in a small vesicle
- A hand lens and needle can be used to remove the mite from a burrow, or the surface of the whole length of a burrow can be removed with a scalpel. The process may be very time-consuming

c) **Treatment**

Preferred drugs
Permethrin 5% cream applied to all areas of the body from the neck down and washed off after 8-14 hours
Alternative drugs
Lindane lotion or cream 1% applied in a thin layer to all areas of the body from the neck down and thoroughly washed off after 8 hours **(Note: do not use in children below 12 years of age, pregnant and lactating women) DO NOT APPLY ABOVE THE NECK.** **OR** Benzyl benzoate 25% solution in adults and 10% or 12.5% in children applied BD for 3 days. Adding a single dose of Ivermectin may improve efficacy in an HIV-positive individual. **DO NOT APPLY ABOVE THE NECK** **OR** 6% Precipitated Sulphur cream or ointment can be applied daily until the patient has improved. It is considered the drug of choice for treatment of scabies in pregnant women and small children

d) Prevention

- Clothes and beddings should be washed normally and pressed with hot iron inside and outside
- Sexual and household contacts should also be treated
- Also avoid touching or sharing clothes with people you know are infected

1.1.20 Proctitis, Proctocolitis and Enteritis associated with STIs

Sexually transmitted gastrointestinal syndromes include proctitis, proctocolitis, and enteritis. Proctitis is inflammation of the rectum. *C. trachomatis*, *N. gonorrhoeae*, HSV, and *T. pallidum* are the most involved sexually transmitted pathogens. Other pathogenic organisms include *Campylobacter sp., Shigella sp*., LGV serovars of *C. trachomatis, and Entamoeba histolytica*. *Cytomegalovirus* (CMV) and other opportunistic infections can be involved in immunosuppressed HIV-infected patients.

a) Clinical features

- Diarrhoea or abdominal cramps
- Anorectal pain
- Tenesmus
- Rectal discharge

b) Diagnosis

- Persons presenting with symptoms of proctitis should undergo an anal examination (Anoscopy if feasible)
- Gram-stained smear of any anorectal exudates from anoscopic or anal examination should be examined for polymorph nuclear leukocytes
- All persons should be evaluated for HSV (PCR or culture)
- *N. gonorrhoeae* (NAAT or culture)
- *C. trachomatis* (NAATs) and
- *T. pallidum* (Dark field microscopy if available and serologic testing)

c) Treatment

If an anorectal exudate is detected on examination, if polymorph nuclear leukocytes are detected on a Gram-stained smear of anorectal exudates or secretions, or if Anoscopy or gram stain is unavailable and the clinical presentation is consistent with sexually acquired acute proctitis, presumptive therapy should be prescribed while awaiting additional laboratory tests.

Preferred drugs
Ceftriaxone 500mg IM stat **OR** Azithromycin 1g PO stat **PLUS** Doxycycline 100mg PO BD for 7 days
Permethrin 1% cream. Apply to clean damp hair. Rinse after 10 minutes and dry

Note:

- Patients with proctitis and a positive rectal Chlamydia NAAT, patients with HIV infection and proctitis, and patients with bloody discharge, perianal or mucosal ulcers should be offered presumptive treatment for LGV with Doxycycline 100mg PO BD for a total of 3 weeks
- All patients with acute proctitis should be tested for HIV if their HIV status is unknown or negative
- Sexual contacts within 60 days before the onset of symptoms should be evaluated, tested and presumptively treated for the respective pathogen
- In HIV-infected persons with acute proctitis who have bloody discharge, painful perianal or mucosal ulcers, presumptive treatment should include a regimen for anogenital herpes and LGV

1.1.21 Sexual Assault And STIs

Victims of sexual assault are likely to contract HIV, other STIs and females may get pregnant. However, among sexually active adults, identification of an STI might represent an infection acquired prior to the assault. Trichomoniasis, gonorrhoea, and chlamydia are the most frequently diagnosed STI among women who have been sexually assaulted. These conditions are relatively prevalent, and the presence after an assault does not necessarily imply acquisition during the assault.

Chlamydial and gonococcal infections in women are of particular concern because of the possibility of ascending infection and the resulting complications. Female survivors who are of reproductive age should also be evaluated for pregnancy.

1.1.22 Evaluating Adolescents and Adults for STIs

a) Initial examinations

- Thorough physical examination should be performed on all survivors
- Decisions on how to approach each patient should be made on an individual basis (considering the nature of the sexual assault)
- Tests for *C. trachomatis*, *N. gonorrhoeae*, and *T. vaginalis* should be done at sites of penetration and on discharge (if present)
- A serum sample should be taken for immediate evaluation of HIV, Hepatitis B, and syphilis

b) Prophylaxis

- Compliance with follow-up visits is poor among survivors of sexual assault. Therefore, the following prophylactic therapy after a sexual assault is recommended:
- An empiric antimicrobial regimen for chlamydia, gonorrhoea, and trichomoniasis
- Emergency contraception if the assault could result in pregnancy in the survivor
- Post-exposure Hepatitis B vaccination (**Note**: refer to National Viral Hepatitis guidelines and seek guidance from the GBV team)
- Link to One stop GBV centre where available
- HPV vaccination (if available) for female and male survivors (give to those who are eligible according to guidelines on vaccination)
- HIV PEP individualized according to risk (See Zambia Consolidated Guidelines for Treatment and Prevention of HIV)

c) Preferred drugs

Preferred drugs
Ceftriaxone 500mg IM stat **PLUS** Azithromycin 1g PO stat **PLUS** Metronidazole 2g PO stat

At the initial examination, sexual assault victims should be counseled regarding

- Symptoms for STIs and the need for immediate examinations if symptoms occur
- Abstinence from sexual intercourse until STI prophylactic treatment is completed

d) Follow-up examinations

After the initial post-assault examination, follow-up examinations provide an opportunity to

- Detect new infections acquired during or after the assault
- Complete Hepatitis B and HPV vaccinations (if indicated)
- Complete counselling and treatment for other STIs (trichomoniasis, chlamydia, gonorrhoea and HIV)
- Monitor side-effects and adherence to PEP medications

If initial testing was done, follow-up examination should be conducted within a week to ensure that results of positive tests are discussed with the patient and treatment is provided if not given at the initial visit.

If the initial test is negative and treatment was not provided, examination for STIs can be repeated within 1-2 weeks of the assault because infectious agents acquired through the assault might not have produced sufficient concentrations of organisms to result in positive test results at the initial examinations.

If treatment was initially provided (with or without testing), testing should be conducted after treatment only if the survivor reports having symptoms. A follow-up examination at 1-2 months should be considered to re-evaluate development of ano-genital warts. Serologic tests for HIV and syphilis can be repeated 6 weeks, 3 months, and 6 months for HIV and 1 and 3 months for syphilis.

OTHER CONDITIONS THAT ARE OF PUBLIC INTEREST

Table 13: Table of Conditions that are of Public Interest

Condition	Causative organism
Monkeypox	*Pox virus*
Zika	*Zika virus*
Lassa Fever	*Lassa virus*
Female Genital Schistosomiasis	*Schistosoma haematobium*
Ebola	*Ebola virus*
Infectious Mononucleosis	*Epstein-Barr virus*

1.1.23 MONKEYPOX

Monkeypox virus was first isolated in 1958 from monkeys at the Staten's Serum Institute in Copenhagen, Denmark—thus its name. However, the natural host of monkeypox virus also includes rope squirrels, tree squirrels, Gambian pouched rats and dormice. As with many zoonoses, this poxvirus is transmitted incidentally to humans when they encounter infected animals. The first known human case of monkeypox was recorded in 1970 in the Democratic Republic of the Congo.

In 2003, monkeypox cases were reported in the US when a shipment of infected Gambian pouched rats subsequently infected prairie dogs housed in the same facility and eventually infected 71 humans who adopted these animals as pets.

The current 2022 outbreak illustrates the easy human-to-human transmissibility by direct, intimate contact with lesions that contain the virus. As of July 2022, more than 3400 laboratory-confirmed cases of monkeypox have been reported from 50 nonendemic countries worldwide, with approximately 86% reported from European region.

a) Clinical presentation

Monkeypox has a wide incubation period ranging from 5 days to 3 weeks. Patients typically present with fever, chills, fatigue, headache, muscle aches, sore throat, lymphadenopathy, and skin lesions. The skin lesions evolve from macules and papules to vesicles and pustules that ulcerate and crust before healing over several weeks. The skin lesions usually occur in crops. The initial lesions are usually at the site of inoculation, which may explain why in the current outbreak lesions have been located on or near the genitalia or anus. Most often, monkeypox infection is self-limited, usually lasting 2 to 4 weeks.

b) Treatment

- Treatment for monkeypox is mostly symptomatic, as there is currently no specific antiviral treatment
- Persons with severe disease, immunocompromised patients, children younger than 8 years, and pregnant individuals should be considered for antiviral treatment: Tecovirimat and Brincidofovir (prodrug of Cidofovir)
- The CDC recommends vaccination within 4 days of exposure to prevent disease or up to 14 days after exposure to reduce severity of disease

Appendices

Annex 1: Common STI Syndromes and their Causative Organisms

Table 14: Common STI Syndromes and their Causative Organisms

STI Syndrome	Disease Entity	Causative Organism
1. **Urethral Discharge**	Gonorrhoea	*N. gonorrhoeae*
	Chlamydia	*C. trachomatis*
	Trichomoniasis	*T. vaginalis*
	Non-specific Urethritis	*U. urealyticum/parvus*
		M. genitalium
		G. vaginalis
2. **Vaginal Discharge**	Gonorrhoea	*N. gonorrhoeae*
	Chlamydia	*C. trachomatis*
	Trichomoniasis	*T. vaginalis*
		U. urealyticum/parvus
		M. genitalium
		G. vaginalis
3. **Genital Ulcer**	Syphilis	*T. pallidum*
	Chancroid	*H. ducreyi*
	Herpes Genitalis	*HSV Type 2*
	Lymphogranuloma Venereum	*C. trachomatis L1-L3*
	Donovanosis	*K. granulomatis*

4. **Inguino Bubo**	Chancroid	*H. ducreyi*
	Lymphogranuloma Venereum	*C. trachomatis L1-L3*
5. **Genital Growth**	Genital warts	HPV
	Condylomata lata	*T. pallidum*
	Molluscum Contagiosum	*Poxvirus*
6. **Scrotal Swelling**	Gonorrhoea	*N. gonorrhoeae*
	Chlamydia	*C. trachomatis*
7. **Female Lower Abdominal Pain**	Gonorrhoea	*N. gonorrhoeae*
	Chlamydia	*C. trachomatis*
8. **Neonatal Conjunctivitis**	Gonorrhoea	*N. gonorrhoeae*
	Chlamydia	*C. trachomatis*
9. **Mixed Syndromes**	*Refer to specific tables*	

ANNEX 2: AVAILABLE DIAGNOSTIC TESTS IN ZAMBIA

Table 15: Current situation of STI testing in Zambia

Aetiological Agent/ Condition	**Recommended Diagnostic Test**	**Current Tests Being Done** Test not done Negative = (-) Test done Positive = (+)				
		Health Post	**Health Centre**	**Hospital L1**	**Hospital L2**	**Hospital L3**
Neisseria Gonorrhoeae (gonorrhoea)	Antigen test (RDT)	-	-	-	-	+
	Smear-Gram stain	-	-	+	+	+
	Culture	-	-	-	+	+
	Organism identification test	-	-	-	+	+
	β-lactamase test	-	-	-	-	+
	Antimicrobial susceptibility test	-	-	-	-	+
	Nucleic acid Amplification (PCR)	-	-	-	-	-
Chlamydia trachomatis	Antigen test (RDT)	-	-	-	-	+
	ELISA	-	-	-	-	-
	Culture	-	-	-	-	+
	Organism identification test	-	-	-	-	+
	Nucleic Acid amplification (PCR)	-	-	-	-	-
Treponema pallidum (syphilis)	Dark field Microscopy	-	-	-	-	-
	RPR Test Qualitative	-	+	+	+	+
	RPR Test Quantitative	-	-	-	+	+
	VDRL (Qualitative)	-	+	+	+	+
	VDRL (Quantitative)	-	-	-	+	+
	TPHA	-	-	-	+	+
	FTA-Abs test	-	-	-	-	-

Aetiological Agent/ Condition	Recommended Diagnostic Test	Current Tests Being Done Test not done Negative = (-) Test done Positive = (+)				
		Health Post	**Health Centre**	**Hospital L1**	**Hospital L2**	**Hospital L3**
	Warthin starry stain	-	-	-	-	+
	Molecular identification (PCR)	-	-	-	-	-
Klebsiella granulomatis	Tissue smear - Giemsa stain	-	-	-	-	-
	Culture	-	-	-	-	-
	Organism identification	-	-	-	-	-
	Antimicrobial Sensitivity	-	-	-	-	-
	Nucleic acid amplification test	-	-	-	-	-
Haemophilus Ducreyi	Smear	-	-	-	-	-
	Culture	-	-	-	-	-
	Organism identification	-	-	-	-	-
	β -lactamase test	-	-	-	-	-
	Antimicrobial susceptibility test	-	-	-	-	-
Herpes Simplex Virus - 1&2	Smear - Tzanck Giemsa stain	-	-	-	-	-
	Immunofluorescent staining	-	-	-	-	-
	ELISA	-	-	-	-	-
	Anti-HSV 1/2 IgM/IgG	-	-	-	-	+
	Specific glycoprotein G [gG]	-	-	-	-	-
	Culture	-	-	-	-	-
	Nucleic Acid amplification	-	-	-	-	-
Trichomonas Vaginalis	Saline wet mount	-	-	-	+	+
	Staining - Giemsa stain	-	-	-	+	+
	Acridine Orange staining	--	-	-	-	-
	Culture	-	-	-	-	-

Aetiological Agent/ Condition	Recommended Diagnostic Test	Current Tests Being Done Test not done Negative = (-) Test done Positive = (+)				
		Health Post	**Health Centre**	**Hospital L1**	**Hospital L2**	**Hospital L3**
	Organism identification	-	-	-	-	-
Bacterial Vaginosis	Saline wet mount					
	Smear - Gram stain					
	pH test, KOH test Whiff test					
	Amstel criteria					
	Oligonucleotide probe					
Human Papilloma Viruses (warts)	Nucleic Acid Amplification	-	-	-	-	+
Hepatitis B Virus	Immunochromatography	-	-	-	-	-
	HBs Ag detection	-	-	-	+	+
	ELISA for HBsAg Detection	-	-	-	-	+
	Nucleic Acid Amplification Tests	-	-	-	-	-
Hepatitis C Virus	Anti-HCV detection	-	-	-	-	-
	Nucleic Acid Amplification Tests	-	-	-	-	-
Mycoplasma Genitalium	Nucleic Acid Amplification Tests	-	-	-	-	-
Candida spp	Potassium hydroxide wet mount	-	+	+	+	+
	Smear - Gram stain	-	+	+	+	+
	Culture	-	-	-	-	+
	Speciation & AST	-	-	-	-	-

ANNEX 3: AVAILABLE DIAGNOSTIC TESTS IN ZAMBIA

List of Examination Equipment

(i). Lights

(ii). Speculum

(iii). Couch

(iv). Screen

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Table 16: List of Reviewers

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Disclaimer:

The reviewers have made every effort to ensure the accuracy and appropriateness of the drug dosages and treatment regimens presented in this booklet. The treatment regimens described are based on best available evidence of STI prevalence and drug sensitivity patterns locally and worldwide. Where local data is lacking or inconclusive, the reviewers have adapted from global STI reviews and treatment guidelines from the WHO and CDC.